

Step 6 Summary — End-to-End Game AI Architectures

Research on the possibilities for applying Artificial Intelligence in computer games

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6 Step 6 — End-to-End Game AI Architectures

6.1 Introduction

Chapter 6 is the keystone of this study plan. Steps 1–5 assembled the parts in isolation — the game-theoretic vocabulary of extensive-form games and Nash equilibria, counterfactual regret minimization (CFR) and its Monte-Carlo variants, game abstraction, and the neural function approximators that replace tabular storage once a game outgrows it — and this chapter is where those parts converge into complete, competition-grade systems. The treatment is deliberately **architectural**: rather than re-deriving the mathematics, which lives in the technical chapters and the cited papers, each system is studied at the level of how it is *put together* — what it computes offline, what it computes in real time, where learning sits, and what it can actually guarantee. A short per-system scorecard opens each section so the same nine dimensions line up at a glance, and the prose then traces the design decisions, the engineering compromises, and the abandoned approaches behind them.

The five systems span seven years and, read in order, trace the evolution of superhuman game AI: **DeepStack** (2017), **Libratus** (2017/2018), **Pluribus** (2019), **ReBeL** (2020), and **Student of Games** (2023). It is tempting to read such a list as a leaderboard — each entry strictly stronger than the last — but that is neither what happened nor how this chapter is organized. The progression is a sequence of *deliberate trades*: each system bought a new capability by giving something else up, and the most instructive moments are the ones where the field stepped sideways or even backward. DeepStack and Libratus are near-contemporaries that proposed *opposite cures* for the same disease — DeepStack discarded the

abstraction-and-blueprint paradigm that had dominated computer poker for nearly two decades, while Libratus kept it and cured its one fatal flaw with real-time, provably safe subgame solving. Pluribus then broke the *player-count* barrier, reaching superhuman six-player play, but only by abandoning safety guarantees altogether and remaining entirely tabular. ReBeL stepped back to two players yet leapt forward on generalization — learning belief-state values with AlphaZero-style self-play, dissolving both the abstraction and the blueprint, and *recovering* the safety Pluribus had surrendered. Student of Games, the capstone, unified perfect- and imperfect-information play in a single algorithm — and paid for that breadth with peak strength, losing decisively to a specialist AlphaZero at Go. Read this way, the chapter is a study of what each advance cost, not a ranking of winners.

Beneath the individual trades, three axes of motion run through all five systems and give the chapter its spine. The first is **representational**: the move from hand-built *abstraction* — bucketing similar hands and allowing only a few bet sizes — toward *learned neural approximation*, in which a network generalizes across situations a table could never enumerate. The second is **temporal**: the move from *solving the entire game offline* into a stored strategy toward *learning a compact model and searching with it in real time*, so that computation is spent at the moment of decision rather than baked in advance. The third axis appears only at the very end, with Student of Games: the unification of **perfect- and imperfect-information** play in one sound algorithm, joining the two great traditions of game AI — the minimax / Monte-Carlo-tree-search / AlphaZero line and the CFR / poker line — that had run on separate tracks for seventy years. The first two axes describe the four poker systems; the third is what makes the fifth the chapter’s capstone.

One idea ties these threads together and deserves to be named before the systems themselves, precisely because it is *not* one of them: **depth-limited solving**. Formalized by Brown & Sandholm (2018), it is the principle that one may search only a little way ahead in an imperfect-information game and substitute a *learned or precomputed value* for the remainder — provided the substitution is done so that hidden information does not render it unsound. It is the theory that retroactively unifies DeepStack’s continual re-solving with Libratus’s nested subgame solving, that underwrites Pluribus’s continuation strategies, and that, in belief-state form, becomes the inner loop of ReBeL and Student of Games. We treat it as connective tissue — referenced wherever a system instantiates it — rather than as a sixth entry, because depth-limited solving is the mechanism *through which* the offline-to-search axis actually operates.

Finally, because this chapter is the hinge between the fundamentals of Steps 1–5 and the opponent-modelling and exploitation work of Steps 7–15, it is worth flagging at the outset the single thread the synthesis returns to. Every system here is, by deliberate design, **opponent-blind**: each computes a strategy that is hard to beat *in the worst case* and then plays it without regard to who is actually across the table — Pluribus does not even know its opponents’ identities, and both it and Libratus refuse on principle to model or adapt to them, so as never to be counter-exploited in return. This robustness-first stance is the field’s great strength and, for a dissertation about *adaptive* play, its defining limitation: it is exactly the opponent-awareness these systems omit that Steps 7–15 set out to add. The chapter therefore closes not with a winner but with a synthesis — a map of what the five systems share, what each gave up, and where the open problems that motivate the rest of this work actually lie.

6.2 DeepStack (2017)

DeepStack (Moravčík et al., 2017), from the University of Alberta computer-poker group with collaborators in Prague, was the first program to defeat professional poker players at heads-up no-limit Texas hold'em (HUNL) with statistical significance, and the first to put heuristic search — the engine behind chess and Go — on a theoretically sound footing in a game of imperfect information. Over a four-week study it beat a pool of 33 professionals by 492 milli-big-blinds per game (mbb/g, the standard poker win-rate unit; 50 mbb/g is a sizable professional edge) across 44,852 hands, and no known technique could find a flaw in its play. Its guiding *what-if* is the one that frames this whole chapter: **could a program play poker the way AlphaGo plays Go — searching locally from the current situation and trusting a learned value function for everything beyond the horizon — even though, in poker, the situation is itself partly hidden?**

At a glance	DeepStack (2017)
Players	2 (heads-up)
Game type	HUNL — heads-up no-limit Texas hold'em (2-player zero-sum)
Blueprint (offline)?	No — offline work trains value nets, not a stored strategy
Neural component	Deep counterfactual value networks (flop, turn, aux); value-only
Search mechanism	Continual re-solving (depth-limited CFR look-ahead, every decision)
Abstraction?	None constrains play; 1,000-bucket clustering only at the net input, plus sparse betting + river action bucketing in look-ahead
Perfect-info too?	No (imperfect-information only)
Compute	Offline-heavy (~175 CPU-core-years to label the turn network); play-time runs on one GPU, < 5 s/decision
Key innovation	Continual re-solving + learned counterfactual values: the first <i>sound</i> heuristic search for imperfect-information games

6.2.1 The gap it closed

Every prior game-AI milestone — backgammon, chess, Go — rested on *local search*: from the current position, look a few moves ahead and substitute a heuristic value for the rest. That recipe assumes the position is known, and poker breaks the assumption — the right action depends on the distribution over the opponent's hidden cards, revealed only through their betting, which in turn depends recursively on what they believe about your cards. The open question DeepStack set out to answer is whether heuristic

search can be made *sound* under hidden information at the scale of HUNL, a game with roughly 10^{160} decision points — comparable to Go, but imperfect-information.

For nearly two decades the dominant answer had been something else entirely: **abstraction plus offline equilibrium plus translation**. Shrink HUNL’s 10^{160} situations into roughly 10^{14} abstract ones by bucketing similar hands (*card abstraction*) and allowing only a few bet sizes (*action abstraction*), solve that smaller game offline with counterfactual regret minimization (CFR) to produce a complete strategy — a *blueprint* — store it, and at play time *translate* each real situation and opponent bet into the nearest abstract one. The compression is lossy, and the loss shows up as exploitability — how much a worst-case opponent can win, the field’s quality metric, zero at a Nash equilibrium. In 2015 the abstraction-based program Claudico lost to professionals by 91 mbb/g, and a local-best-response probe (LBR, a tractable lower bound on exploitability) later showed top competition bots exploitable by more than 3,000 mbb/g — four times worse than folding every hand. Blueprints were also enormous (a single no-card-abstraction strategy took about 2 TB and 14 CPU-years to compute and was *still* exploitable through off-tree bets), and the translation step was itself a source of weakness. DeepStack closes this gap by discarding the whole edifice: it never builds a full-game abstraction and never stores a blueprint, reasoning about each situation *as it actually arises* and replacing only the distant remainder of the game with a learned estimate.

6.2.2 Architecture

DeepStack splits cleanly into an **offline** phase that learns intuition and an **online** phase that searches with it (Figure 6.1).

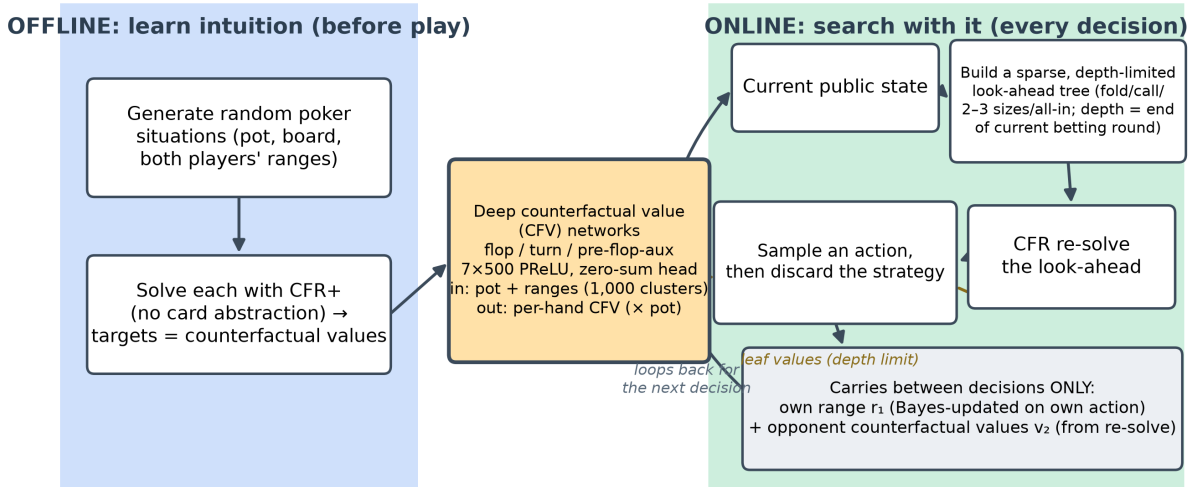


Figure 1: DeepStack architecture: offline intuition-building (left) feeding a shared deep counterfactual-value network, reused as the leaf evaluator inside the online continual-re-solving loop (right).

Offline, the system generates millions of random poker situations and solves them with a CFR solver to obtain target *counterfactual values* — conditional “what-if” payoffs for holding each possible hand — and these (situation → value) pairs train the **deep counterfactual value (CFV) networks**. Online, DeepStack carries no blueprint: between decisions it remembers only two vectors — its own *range* (the

distribution over the hands it could be holding) and the opponent’s *counterfactual values* — and at each turn it runs CFR over a small look-ahead tree rooted at the true current state, using a CFV network to supply leaf values at the depth limit. CFR already computes both ranges and counterfactual values, which is why it slots so naturally into this loop.

This is heuristic search with three ingredients: a sound local strategy computation (continual re-solving), depth-limited look-ahead with a learned value function in place of the rest of the game, and a restricted set of look-ahead actions to keep each solve fast. The neural network sits *only* at the depth limit as a leaf evaluator; CFR does the searching; and — crucially — there is no full-game abstraction in the loop.

6.2.3 Key innovation: continual re-solving with learned counterfactual values

DeepStack’s contribution is the marriage of two ideas, each of which makes the other practical.

The first is **continual re-solving**: reconstructing a fresh local strategy at every decision from the maintained range and opponent counterfactual-value vector, then discarding it, so the agent never stores or commits to a global strategy. Classical re-solving (Burch et al., 2014) shows that to reconstruct a strategy for a subgame you do not need the whole strategy — you need only your own range entering the subgame and a vector of the opponent’s counterfactual values. DeepStack pushes this to its limit: it *never* holds a strategy for the whole game. Every time it must act, it re-solves the current public state from those two vectors, plays one action, and throws the strategy away.

What makes this work is the bookkeeping. After each event DeepStack updates its two vectors by simple rules: on its **own action** it swaps in the re-solved counterfactual values for the chosen action and Bayes-updates its range; on a **chance event** it swaps in that card’s values and zeroes the now-impossible hands; and on the **opponent’s action** it does *nothing at all*. That last point is the quiet masterstroke: because it tracks the opponent’s *values* rather than their *range*, and never needs the opponent’s specific action to maintain those values, it sidesteps the action-translation step that crippled abstraction-based bots. The opponent can make any bet of any size; DeepStack simply re-solves from the state that bet produced.

The second idea is the **learned counterfactual value network** — DeepStack’s “intuition.” In a perfect-information game a leaf evaluator maps one state to one number; under imperfect information it must map a *whole public state plus both players’ ranges* to a *vector* of counterfactual values, one per hand, because the values at a node shift with the ranges that reach it. DeepStack learns this map with a feed-forward network of seven hidden layers of 500 units, taking the pot size and the two ranges (compressed into 1,000 hand clusters) as input and emitting per-hand counterfactual values as fractions of the pot. A bespoke outer layer enforces the zero-sum constraint: it forms the two implied game values from the ranges and raw outputs and subtracts half their sum, so the estimates are mutually consistent and the whole thing stays differentiable. With this network supplying values at the end of the current betting round, the depth-limited re-solve shrinks the game from 10^{160} decision points to about 10^7 — small enough to solve in under five seconds on a single GPU.

The pairing is provably sound. If the value network’s error is at most ϵ and the re-solve runs T CFR iterations, the resulting strategy’s exploitability is bounded by

$$\text{exploitability} < k_1 \epsilon + k_2 / \sqrt{T},$$

with game-specific constants k_1, k_2 . The first term is the price of imperfect intuition; the second is ordinary CFR convergence. This bound is the theoretical heart of the paper — the guarantee that heuristic search can be carried into imperfect information without the strategy quietly becoming exploitable — and the same depth-limit-plus-learned-value template, formalized further by Brown & Sandholm’s depth-limited solving the following year, underlies every system in this chapter.

6.2.4 Caveats, dead-ends, and what the paper under-describes

The clean story above is the main text; the engineering reality lives in the supplement, and it is where the spine’s “caveats and evolution” emphasis earns its keep.

The most important asterisk is that **the deployed system is not the one the theorem covers**. To play at human speed DeepStack restricts its look-ahead to a sparse betting set (fold, call, two or three bet sizes, all-in), and the paper states plainly that this “voids the soundness property of Theorem 1.” So the shipped guarantee is empirical — supported by the LBR results below — not proven. A second deviation compounds this: the soundness proof assumes *best-response* constraint values, but DeepStack actually uses *self-play* values, which “lack a theoretical justification” yet were less exploitable in early tests. The proven algorithm and the winning algorithm are, strictly, different algorithms.

Abstraction also creeps back at the margins. DeepStack advertises that it uses no card abstraction to constrain play — but it *does* cluster hands into 1,000 buckets at the value network’s input, and on the river it abandons the network entirely, solving to the end of the game while using a **bucketed action abstraction** for tractability. To speed up re-solving it also **warm-starts** the opponent’s range, in a conservative blend most of the time but in an aggressive variant (when acting first) that “sacrifices the re-solving guarantees when the opponent’s range estimate is wrong.” And the pre-flop, far from cheap, requires enumerating all 22,100 possible flops through the flop network, mitigated only by caching repeated betting sequences. Finally, the choice of re-solving gadget (the CFR-D gadget over the max-margin alternative) was made because it “performed better in early testing,” not derived — and, echoing Step 5, the network’s useful depth was capped by data rather than architecture, with validation error flattening past five layers at the ten-million-sample budget. None of these undermine the result, but together they show that the headline “sound, abstraction-free search” is an aspiration the implementation approximates rather than attains.

6.2.5 Compute & accessibility

DeepStack’s cost is almost entirely **offline, and almost entirely in the CFR solving used to label the value networks** — not in the neural training, and not in play. Generating the targets meant approximately solving millions of random subgames: the turn network alone consumed about **175 CPU-core-years** on a 6,144-core cluster, with the flop network adding roughly half a GPU-year on 20 GPUs; by contrast, the networks themselves trained in about two days each on a single GPU, and at play time

DeepStack runs on **one commodity GPU at under five seconds per decision**. The shape of that bill is the real lesson: the “intuition” is bought once, up front, by brute-force equilibrium solving, after which deployment is cheap — the mirror image of a system whose cost is dominated by inference. In accessibility terms this put a *from-scratch* build within reach only of a well-resourced lab at the time (the offline solve is cluster-scale), even though the trained agent then ran on hardware any enthusiast owned; a decade on, that offline cost is far more tractable on commodity cloud compute and with faster modern solvers, and a public reference implementation on Leduc lowers the entry barrier further.

6.2.6 Strengths and limitations

DeepStack’s central strength is **soundness with low exploitability**: it is the first imperfect-information search method with a real guarantee, and empirically LBR — which exposes competition bots as losing thousands of mbb/g — cannot find any way to beat it, itself losing by over 350 mbb/g. It needs **no full-game abstraction and no action translation**, so off-tree opponent bets are handled exactly rather than rounded; its **play-time footprint is modest**; and it learns from **no human data and little domain knowledge**. A neat bonus is evaluation synergy — DeepStack’s own value function is exactly what the AIVAT variance-reduction estimator needs, cutting the human-study standard deviation by 85% and making significance achievable in only a few thousand hands.

The limitations set the agenda for the rest of the chapter. DeepStack is **two-player zero-sum only**; its soundness leans on that structure. It retains a **sparse betting abstraction** in the look-ahead and an action abstraction on the river, and it inherits **value-network approximation error** (the $k_1\epsilon$ term), worst on the flop where it leans hardest on the network. It pays the **heavy offline cost** noted above and **re-solves from scratch** at every decision, and it was validated only against humans and LBR — never head-to-head against the strongest abstraction bots.

6.2.7 Legacy and modern relevance

Strip away the poker specifics and DeepStack’s core idea is **depth-limited search with a learned value function at the leaves, adapted to hidden information** — the imperfect-information counterpart of the value-guided search behind AlphaGo and AlphaZero. That idea did not age into obsolescence; it became the template. Brown & Sandholm formalized the depth-limited-solving theory the next year, and the paradigm was then generalized into full reinforcement-learning-plus-search frameworks: ReBeL (2020) recast the leaf-value learning around public belief states with AlphaZero-style self-play, and Student of Games (2023) — which shares several DeepStack authors — unified perfect- and imperfect-information play in a single algorithm. The same “planning in the loop with a learned value model” recipe later reached beyond two-player zero-sum poker, for instance in CICERO’s human-level Diplomacy play. More broadly still, DeepStack is an early, clean instance of the principle now central to frontier AI: **spend compute at decision time via search guided by learned intuition, rather than baking everything into one giant precomputed policy** — the same “test-time compute” thesis behind today’s reasoning models, with the field’s own caveat (voiced by Noam Brown) that imperfect information needed *belief-aware* search precisely because the plain Monte-Carlo tree search that works for Go does not work for poker.

Several concrete subsystems remain reusable today: the depth-limit-plus-learned-leaf-value pattern itself; the differentiable **zero-sum-consistency output head** for two-player value networks; **AIVAT-style variance reduction**, which turns a learned value model into a control variate for low-variance evaluation of any stochastic agent; and the small but transferable trick of expressing values as **fractions of the pot** for scale-invariant generalization. What is genuinely superseded is the hand-engineered machinery *around* the idea — the 1,000-bucket k-means clustering, the separate per-round networks, the hand-tuned re-solving gadget — which the more general learned representations of ReBeL and Student of Games replace. The honest verdict: **as a deployed poker system DeepStack is superseded, but it is far from a mere stepping stone** — its central paradigm won and is now mainstream, and for this thesis specifically its (range, opponent counterfactual values) state and its error-propagation bound are direct seeds for belief-based opponent modelling (Contribution 1) and safe-exploitation analysis (Contribution 2).

6.3 Libratus (2017/2018)

DeepStack answered its *what-if* by throwing the abstraction-and-blueprint edifice away — yet it still kept a sparse betting abstraction inside its own look-ahead, and it was never tested head-to-head against the strongest prior bots or against HUNL specialists in a long, rigorous match. Libratus (Brown & Sandholm, 2017; Carnegie Mellon) — Latin for *balanced*, as in approximating a Nash equilibrium, and *forceful*, for its play — was built independently and announced the same year, and it made the opposite bet: keep the abstraction-and-blueprint paradigm and cure its one fatal disease. Its guiding question is the mirror image of DeepStack’s: **what if we solve a coarse blueprint of the whole game offline, then *repair it in real time* wherever the abstraction is too crude — with a provable guarantee that the repair never leaves us more exploitable?** In January 2017 Libratus became the first program to beat top human HUNL specialists in a long match, defeating four professionals by 147 mbb/g (milli-big-blinds per game, the win-rate unit from the previous section; ~50 mbb/g is a sizable professional edge) over 120,000 hands at 99.98% significance — and, unlike DeepStack, it first dismantled the prior best poker AI head-to-head.

At a glance	Libratus (2017/2018)
Players	2 (heads-up)
Game type	HUNL — heads-up no-limit Texas hold’em (2-player zero-sum)
Blueprint (offline)?	Yes — an abstracted full-game strategy solved offline with MCCFR (detailed early, coarse late)
Neural component	None — purely tabular CFR / abstraction (no neural networks anywhere)
Search mechanism	Nested safe subgame solving (real-time CFR+ re-solve of the late game, re-run for every off-tree opponent bet)

At a glance	Libratus (2017/2018)
Abstraction?	Yes — card abstraction (turn/river, blueprint only) + asymmetric action abstraction; dissolved to <i>no card abstraction</i> inside the real-time subgames
Perfect-info too?	No (imperfect-information only)
Compute	Offline- <i>and</i> online-heavy: ~25M CPU core-hours on the Bridges supercomputer; ~50 nodes and tens of seconds per late decision; no GPUs
Key innovation	Blueprint + real-time nested <i>safe</i> subgame solving + self-improvement: exact, provably-safe responses to off-tree bets in place of action translation

6.3.1 The gap it closed

The previous section laid out the paradigm DeepStack discarded — abstraction plus offline equilibrium plus translation — and named its fatal flaw. Libratus targets exactly that flaw without discarding the paradigm. The blueprint is solved in advance over a *fixed* menu of bet sizes; at play time any opponent bet that is not on that menu is *translated* — rounded to the nearest size the bot knows — after which the bot replies as though the opponent had made the rounded bet. That rounding is the single largest exploitable seam in abstraction-based poker: a local-best-response probe had shown the leading competition bots losing thousands of mbb/g to a worst-case adversary, and in 2015 Libratus’s own predecessor Claudico lost the first *Brains vs. AI* match to professionals by 91 mbb/g, in good part because opponents could feel out and punish its translation boundaries.

The conceptual question Libratus answers is therefore narrower and more surgical than DeepStack’s: *can the decades-old abstraction paradigm be made superhuman by repairing only its real-time behaviour — responding to off-menu bets exactly rather than by rounding — and can that repair be done with a guarantee that local fixes never increase global exploitability?* The answer is the three-module architecture below. Where DeepStack removed the blueprint entirely and leaned on a learned value function, Libratus keeps the blueprint as a cheap scaffold and adds a real-time solver that, whenever the opponent steps off the tree, builds and solves a finer subgame *containing the actual bet* and stitches the result back into the blueprint. The same year, then, produced two opposite cures for the same disease — one neural and blueprint-free, one tabular and blueprint-based — and Libratus is the proof that the older paradigm, properly repaired, was still enough to reach superhuman play.

6.3.2 Architecture

Libratus is a pipeline of three modules that operate on three different timescales — **offline** (before the match), **online** (during each decision), and **overnight** (between days of play) — and, unlike every other system in this chapter, it contains **no neural network at all** (Figure 6.2).

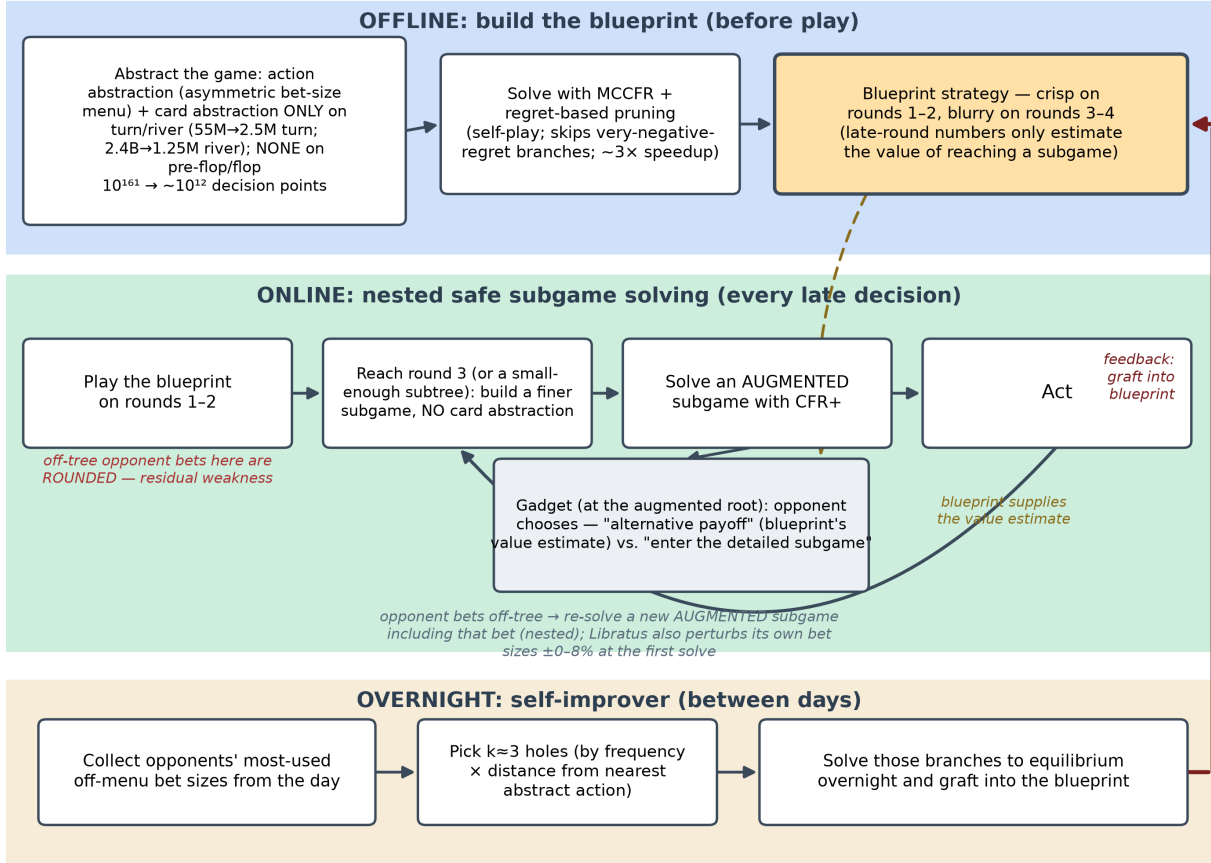


Figure 2: Libratus's three-module pipeline: an offline blueprint (top), the online nested safe-subgame solver that plays it (middle), and the overnight self-improver that grafts solved holes back into the blueprint (bottom, feedback arrow).

Module 1 — the blueprint (offline). Libratus first compresses HUNL’s roughly 10^{161} decision points to about 10^{12} with two kinds of abstraction: an *action abstraction* that keeps only a discrete menu of bet sizes (mostly round fractions and multiples of the pot, drawn from the sizes top competition bots favour, with a few early sizes tuned by a parameter-optimization algorithm), and a *card abstraction* that groups strategically similar hands. Crucially it uses **no card abstraction on the first two betting rounds** — small enough to afford full resolution — and buckets only the turn and river, and even there only in the blueprint. It then solves this abstract game by self-play with an improved **Monte Carlo counterfactual regret minimization (MCCFR)** that probabilistically prunes very-negative-regret branches, a roughly three-fold speedup that also quietly heals a pathology of imperfect-recall abstractions, in which several distinct situations share one bucket’s strategy and “fight” over it. The result is the **blueprint**: a complete but uneven strategy — detailed early, coarse late — whose late-round numbers are used not to play but only to *estimate the value of reaching a subgame*.

Module 2 — nested safe subgame solving (online). Libratus plays the blueprint only in the early rounds. On reaching the third betting round — or any earlier point where the rest of the hand is small enough — it discards the coarse late-game blueprint and instead **builds a fresh, finer-grained subgame with no card abstraction and solves it in real time** with a heavily optimized **CFR+** (a fast, deterministic CFR variant). Each hand is thus played individually in the late game. The defining feature is what happens when the opponent makes a bet that is not on the blueprint’s menu: rather than round it, Libratus solves a new subgame that *contains that exact bet*, and repeats this for every subsequent off-menu action — *nested* subgame solving. This is the module that the next subsection dissects.

Module 3 — the self-improver (overnight). Because real-time solving is skipped on the first two rounds (solving so early would itself demand heavy abstraction), off-menu opponent bets *there* are still rounded. The self-improver narrows this residual seam between days: it tallies the opponents’ most-used off-menu bet sizes, chooses a few of the most damaging holes, computes proper game-theoretic responses to them overnight, and grafts those branches into the blueprint. Pointedly, this is **not opponent exploitation** — it never tries to model and punish the humans’ mistakes (which would expose Libratus to counter-exploitation). Instead it uses the opponents’ bets only as a hint about *which of Libratus’s own holes to patch*, and the patches are universal, improving play against any future opponent.

The classic building blocks of earlier steps sit in plain view: game abstraction and MCCFR build the blueprint offline, CFR+ does the online subgame solving, and “search” is the per-decision real-time resolve. The conspicuous absence — no value network, no policy network, nothing learned by gradient descent — is the architectural signature that separates Libratus from every other system in this chapter.

6.3.3 Key innovation: nested safe subgame solving

Libratus’s central contribution is making **real-time subgame solving both *safe* and *nested*** — strong enough to beat humans and provably unable to make the strategy much more exploitable than the blueprint it refines. Both adjectives matter, and each fixes a specific prior failure.

The difficulty is that an imperfect-information subgame *cannot be solved in isolation*. In chess, once you

reach an endgame you can solve it on its own, because the board is fully known. In poker the right strategy for a subgame depends on the strategies in *other, unreached* subgames, because those determine the probability distribution over the hidden hands the opponent could hold here. Earlier real-time solvers sidestepped this by assuming the opponent had played the blueprint up to this point — **unsafe** subgame solving — which the opponent can punish simply by deviating, sometimes producing a refined strategy far *worse* than the blueprint.

Libratus’s **safe** subgame solving removes that assumption with a small gadget, the *augmented subgame*. At the root of the subgame the opponent is handed a choice for every hand they might hold: either take a fixed **alternative payoff** — the blueprint’s estimate of what that hand is worth here — or *enter* the detailed subgame and play it out. Solving this augmented game forces Libratus’s refined strategy to make the opponent **no better off than that estimate for every possible hand**, which is exactly the meaning of “safe”: the local repair cannot be exploited relative to the blueprint, whatever the opponent does. Libratus sharpens this with a variant the authors call **Estimated-Maxmargin** — it maximizes the *smallest* safety margin across all opponent hands — and with two refinements that make it strong rather than merely safe: it uses blueprint *estimates* of opponent values instead of conservative *upper bounds* (less timid play), and it *de-emphasizes hands the opponent could only hold by having made an earlier mistake* (it is pointless to defend against a hand a rational opponent would never have brought to this spot), freeing capacity to defend against the hands they realistically have.

The pairing comes with a guarantee that parallels DeepStack’s. If σ^* is the least-exploitable strategy that differs from the blueprint only inside the solved subgames, and the blueprint’s estimate of the opponent’s subgame values is off by at most Δ , then the refined strategy’s exploitability obeys

$$\text{exploitability}(\sigma_{\text{refined}}) \leq \text{exploitability}(\sigma^*) + 2\Delta.$$

The 2Δ term is the price of imperfect value *estimates*, and it is the structural twin of DeepStack’s $k_1\epsilon$ term — both bound how much an error in the values handed to the solver can cost in worst-case exploitability. The two systems thus arrive at the same kind of soundness statement from opposite directions: DeepStack’s ϵ is a *neural network’s* approximation error at the depth limit; Libratus’s Δ is a *tabular blueprint’s* estimation error at the subgame boundary.

“Nested” is the second half. Rather than expand the blueprint’s bet menu (which would balloon the offline solve), Libratus crafts a **distinct response to each off-menu bet in real time**: when the opponent bets a size it has never enumerated, it builds an augmented subgame whose alternative payoff is the best *in-menu* action the opponent could have taken instead, solves it, and does so afresh for every later off-menu bet down the hand. In the controlled experiments this beat the previous standard — rounding the bet to the nearest menu size — by more than an order of magnitude in worst-case exploitability (119 versus 1465 mbb/g in the reported small-game test). And because each subgame is solved live, Libratus can also *change its own bet sizes* between hands — it perturbs them by a random 0–8% at the first solve — so the humans never face a fixed target to dissect. This is the precise mechanism by which the abstraction

paradigm’s worst weakness, action translation, is excised from the late game.

6.3.4 Caveats, dead-ends, and what the paper under-describes

As with DeepStack, the clean three-module story hides where the engineering — and the honesty about limitations — actually lives, except that here much of it is not even in the *Science* paper but in the companion IJCAI paper and the authors’ course notes.

The most revealing caveat is that **the blueprint alone is not superhuman — it does not even beat the prior bot**. Against Baby Tartanian8, the 2016 competition winner, Libratus’s raw blueprint *lost* by 8 mbb/g; only when nested subgame solving was switched on did the same system win by 63 mbb/g. The offline strategy, in other words, is a scaffold, and essentially all of Libratus’s edge comes from real-time search — a result that quietly reframes the whole system and foreshadows the field’s pivot toward search-at-inference.

Several other asterisks matter. **Action translation is not eliminated, only contained**: on the first two betting rounds Libratus still rounds off-menu opponent bets, and the entire self-improver module exists to chip away at that residual seam, three holes per night, never closing it. **Safe subgame solving had to be re-engineered to be usable at all** — for three years it was considered impractical because, in head-to-head play, textbook-safe solving lost to the theoretically unjustified *unsafe* variant; Libratus’s use of value *estimates* rather than conservative *bounds*, together with its de-emphasis of hands an opponent would only hold by mistake, is what made safety competitive, and even so Libratus deliberately uses *unsafe* solving once, at its first entry into the third round, because it is cheaper and empirically fine there. The **Estimated-Maxmargin** choice itself trades a little theoretical purity for strength: using estimates rather than upper bounds can in principle push exploitability *above* the blueprint’s, bounded only by the 2Δ of the theorem. And the imperfect-recall abstraction underneath it all means several distinct situations share a strategy and “fight” over it — a known distortion the regret-based pruning only partly mitigates. Finally, the **compute is almost absent from the main paper**: the headline article gives essentially no figures, and the true cost (next) surfaces only in the secondary sources, as does the fact that the **code was never released**, leaving independent verification to rest on the published pseudocode.

6.3.5 Compute & accessibility

Where DeepStack’s bill is paid almost entirely offline and its play is cheap, Libratus is **expensive at both ends**. The project consumed roughly **25 million CPU core-hours** on the Bridges supercomputer at the Pittsburgh Supercomputing Center over a year — of which about 6 million went to building and solving the blueprint, about 3 million to real-time subgame solving during the match, about 3 million to the self-improver, and the remaining ~13 million to exploratory experiments and evaluation. Operationally the blueprint runs occupied roughly 195 nodes for one to eight weeks at a time; each **real-time subgame solve used about 50 nodes and took on the order of tens of seconds**; the overnight self-improver ran on up to several hundred nodes for hours; and the strategies and snapshots consumed about 2.6 petabytes of disk. There were **no GPUs and no neural training anywhere** — every core-hour is CFR or abstraction.

The shape of that bill is the lesson. Libratus does not buy cheap deployment with an expensive one-time training run, the way DeepStack does; its *play-time* cost is itself supercomputer-scale, because the strength comes from solving a fresh subgame at almost every late decision. In accessibility terms this made Libratus, as deployed, essentially impossible to reproduce outside a major supercomputing centre at *both* training and play time — a far higher bar than DeepStack’s single-GPU agent — and the absence of any code release reinforced that. The underlying techniques are application-independent and run far more cheaply on modern hardware and solvers today, but the 2017 artefact was a demonstration of what was possible with abundant compute, not of what was broadly attainable.

6.3.6 Strengths and limitations

Libratus’s signal strength is **decisive, rigorously demonstrated superhuman play**. It is the first AI to beat top HUNL *specialists* in a long match — 147 mbb/g over 120,000 hands at 99.98% significance, beating each of the four professionals individually — and, unlike DeepStack, it also beat the **prior best poker AI head-to-head** by 63 mbb/g, disentangling its strength from the question of human skill. Its core technique, nested safe subgame solving, is both **provably safe** (the 2Δ bound) and **empirically an order of magnitude less exploitable** than the action translation it replaces, and it uses **no human data and no domain-specific knowledge**. The modular design is itself a contribution: a cheap blueprint, a real-time solver that repairs it where it matters, and an overnight loop that patches its own holes — each module covering another’s weakness — and the whole system is **robustness-first**, refusing to exploit opponents so as never to be counter-exploited.

The limitations are equally clear and set up the rest of the chapter. Libratus is **two-player zero-sum only**; its safety guarantees lean on that structure. It remains **abstraction-based**, and still **translates off-menu bets on the early rounds**, an exploitable seam the self-improver only narrows. It uses **no neural generalization** whatsoever: nothing transfers across situations, the blueprint is enormous (petabytes of stored strategy), and play time is **supercomputer-scale** rather than the single GPU DeepStack needed. The Estimated-Maxmargin trade-off accepts a small, bounded rise in exploitability for strength, one unsafe solve slips into the pipeline, and — because both humans and AI adapted over the match — even the headline significance is, strictly, an “as-if-independent” figure, though a 147 mbb/g margin over 120,000 hands leaves no real doubt.

6.3.7 Legacy and modern relevance

Strip away the abstraction machinery and Libratus’s enduring idea is **real-time search layered on a coarse precomputed strategy, made safe under hidden information**: solve the local situation exactly at decision time, consistently with a cheap global plan, and respond to whatever the opponent actually does rather than to a rounded approximation of it. That idea did not just survive — it became, with DeepStack’s continual re-solving, half of the foundation the rest of the chapter is built on. Libratus’s nested safe subgame solving is the direct ancestor of the real-time search in **Pluribus** (2019), it is unified with DeepStack’s re-solving by the **depth-limited solving** theory the authors published the following year, and the same “refine a global value estimate with a local solve” pattern reappears, now with *learned*

values replacing the tabular blueprint, in **ReBeL** (2020).

Libratus is also, in a precise sense, the **high-water mark of a paradigm that the field then left behind**. It reached superhuman play with **no deep learning at all** — pure abstraction and CFR — and yet the subsequent systems are all neural, because hand-crafted abstraction and petabyte blueprints neither generalize across situations nor scale beyond two players. The honest verdict is that Libratus is *superseded as an architecture but vindicated as a thesis*: its bet that **real-time search matters more than a bigger precomputed strategy** was exactly right, even as its bet on tabular abstraction was overtaken. That first thesis has since become a central theme of frontier AI — the observation that adding search at decision time was worth far more than scaling the offline computation is an early, concrete instance of the “test-time compute” argument now made for reasoning models, and Libratus’s blueprint-then-search split prefigures the modern pretrain-then-search recipe.

Several subsystems remain directly reusable: **safe subgame / endgame solving** as a way to locally refine a global policy *without breaking its guarantees*; the **augmented-subgame gadget** — the clean “take a known alternative value or enter the subgame” construction for pinning a local solution to a global value estimate; **regret-based pruning** for scaling CFR; and the **self-improver pattern** of using observed play to find and *fix one’s own holes* rather than to exploit the opponent. What is genuinely obsolete is the surrounding edifice — the hand-tuned card and action abstraction, the multi-petabyte tabular blueprint, and the cluster-at-play-time deployment — all of which the learned representations of later systems dissolve. For this thesis specifically, Libratus contributes two seeds: its safe-subgame-solving exploitability bound is a template for **safe exploitation under value error (Contribution 2)**, and its deliberate refusal to exploit — “fix your own weaknesses, do not model the opponent” — is the precise foil against which a *bounded, deliberate* opponent adaptation (Contribution 1) can be defined.

6.4 Pluribus (2019)

Libratus settled two-player no-limit hold’em, but its safety guarantees — and indeed the very meaning of “solving” the game — rested on two-player zero-sum structure: there a Nash equilibrium is unbeatable, and a real-time solve could be made provably *safe* against it. Poker as humans actually play it, though, seats six. Pluribus (Brown & Sandholm, 2019; Carnegie Mellon and Facebook AI) confronted the multiplayer question directly: **what becomes of the blueprint-plus-real-time-search recipe when you remove the two-player crutch and sit at a six-handed table — a setting where a Nash equilibrium is neither unique, nor efficiently computable, nor even a guarantee that you will not lose?** Its answer was empirical and emphatic. Across two formats — five professionals seated with one copy of Pluribus, and one professional against five copies — it beat a rotating cast of thirteen elite pros, several of them World Series or World Poker Tour champions, winning by about **48 milli-big-blinds per game** against five humans at once (mbb/g, the win-rate unit from the previous sections — roughly five big blinds per hundred hands, a decisive six-handed margin) at 95% statistical significance. And it did so after training for **eight days on a single 64-core server for about \$150 of cloud compute** — on the order of a thousandth of what the supercomputer behind Libratus consumed.

At a glance	Pluribus (2019)
Players	6 (six-max) — the first superhuman AI in any benchmark game with more than two players/teams
Game type	6-max NLHE — six-player no-limit Texas hold'em (imperfect-information; multiplayer, <i>not</i> 2-player zero-sum)
Blueprint (offline)?	Yes — a full-game blueprint solved offline by Linear MCCFR; played <i>directly</i> only on the first betting round, a scaffold thereafter
Neural component	None — purely tabular CFR / abstraction (no neural networks anywhere, as in Libratus)
Search mechanism	Real-time <i>depth-limited</i> search with $k=4$ continuation strategies at the leaves (nested <i>unsafe</i> solving, re-solved from the start of each betting round); Linear CFR inside subgames
Abstraction?	Yes — action (1–14 blueprint bet sizes; 1–6 in search) + information abstraction (lossless first round; lossy buckets later, finer in search than in the blueprint)
Perfect-info too?	No (imperfect-information only)
Compute	Famously cheap: blueprint ~12,400 core-hours / 8 days / < 512 GB on one 64-core server (~\$144); play on 2 CPUs (28 cores), < 128 GB, no GPUs, 1–33 s/decision
Key innovation	Superhuman six-player play via blueprint + depth-limited search with continuation strategies — won <i>empirically</i> , on a tiny budget, with no N-player safety guarantee

6.4.1 The gap it closed

Every superhuman game AI before Pluribus — checkers, chess, Go, and both prior poker programs — shared a hidden assumption: two players, zero sum. In that setting a Nash equilibrium carries a property that makes it the obvious target: any player who adopts one is *guaranteed not to lose* in expectation, whatever the opponent does, and if two players independently compute equilibria, their strategies still combine into an equilibrium. “Solving” such a game therefore *means* approximating a Nash equilibrium, and DeepStack and Libratus were, at bottom, two ways to do exactly that. Six-handed poker dissolves all of it. With more than two players, finding — even approximating — a Nash equilibrium is computationally intractable in general; there are typically *many* equilibria; and, fatally, if each player independently selects

one, the resulting joint strategy need not be an equilibrium at all, so a player can *lose* while playing an impeccable equilibrium strategy. Brown and Sandholm illustrate this with the Lemonade Stand Game, in which players space themselves around a ring: there are infinitely many equilibria, and independently chosen ones rarely mesh. The conclusion is stark — in multiplayer poker a Nash equilibrium is *neither unique nor a safety guarantee*, and the field’s two-decade definition of success simply evaporates.

Pluribus closes this gap by changing the goal and then meeting it. Rather than chase a solution concept it cannot trust, it aims only to *empirically and consistently defeat elite humans*, and it accepts up front that its algorithms carry no guarantee of converging to an equilibrium outside two-player zero-sum play. That concession is the entire point — and it is the precise gap this thesis’s Contribution 2 sets out to fill. The other half of the gap is mechanical. Libratus reached superhuman play by solving every late-game subgame *to the end* in real time; with six players the subgames explode exponentially and solving to the end becomes infeasible, while DeepStack’s alternative — leaf values conditioned on belief distributions — was already costly in two-player poker and worsens as players multiply. Pluribus therefore needed a real-time search that looks only a little way ahead and stops — in a game where, as the next sections show, stopping early is exactly what naïve search cannot safely do.

6.4.2 Architecture

Like Libratus, Pluribus splits into an offline phase that builds a blueprint and an online phase that searches — but the balance of power between them is inverted, and, again like Libratus, there is **no neural network anywhere** (Figure 6.3).

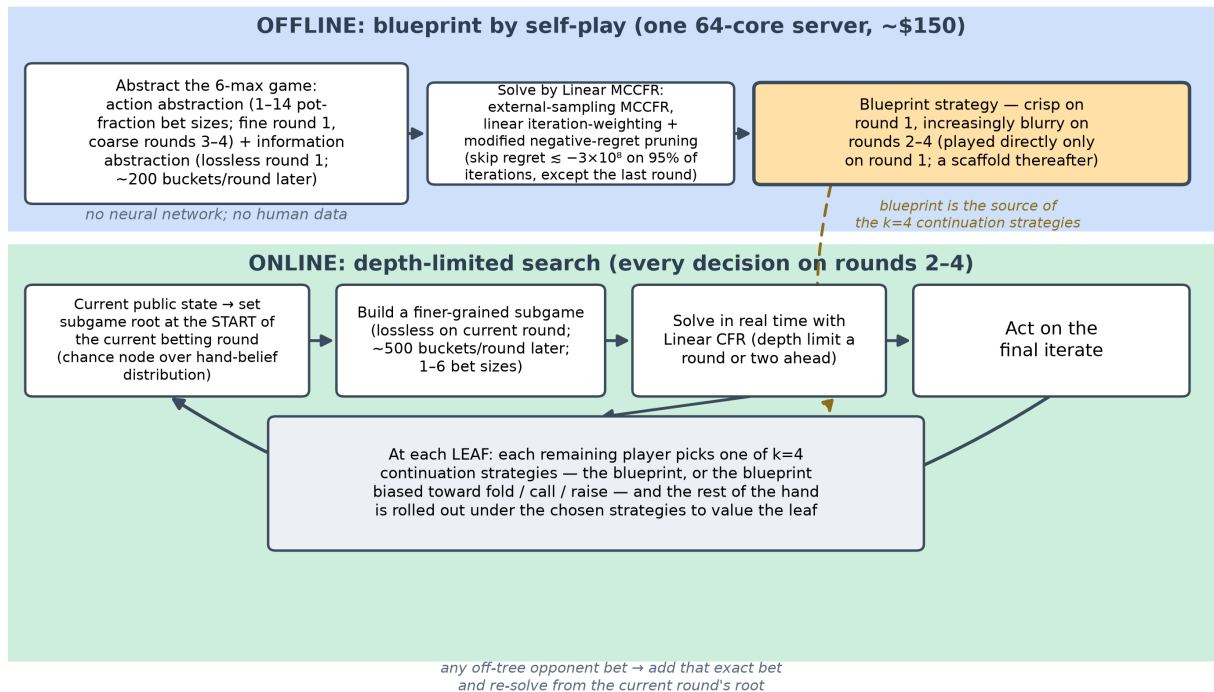


Figure 3: Pluribus’s two-phase architecture: an offline Linear-MCCFR blueprint (top) reused as the source of $k=4$ continuation strategies inside the online depth-limited search (bottom).

Offline, Pluribus computes a blueprint for the whole game by self-play, starting from random play and improving through an **external-sampling Monte Carlo CFR (MCCFR)** that, each iteration, picks one player as the “traverser,” simulates a hand, and nudges that player toward the actions that did better. It abstracts the game two ways — a handful of bet sizes (*action abstraction*) and buckets of strategically similar card situations (*information abstraction*) — but only coarsely, and it plays this blueprint *directly* in just the first of the four betting rounds, where the action abstraction is kept fine and no card bucketing is used. Everywhere else — flop, turn, river — Pluribus discards the coarse blueprint locally and **searches in real time** for a finer strategy, exactly as Libratus did, with one decisive difference: it does not solve to the end of the game. It looks only a round or two ahead to a depth limit and stops, which is what makes six players affordable.

The familiar building blocks are all in view — abstraction and MCCFR build the blueprint, Linear CFR does the online solving, and “search” is the per-decision re-solve — and the two architectural signatures are both *absences*. There is **no neural network** (the system is purely tabular, like Libratus), and, unlike Libratus, there is **no self-improver**: Pluribus never patches its blueprint between sessions, because depth-limited search on three of the four rounds already does the repairing. Two ingredients make stopping early sound, and they are the subject of the deep dive below: a careful answer to what a “leaf value” can even mean under hidden information, and a search that begins not at the current decision but at the *start* of the current betting round.

6.4.3 Key innovation: depth-limited search with continuation strategies

Heuristic search in chess or Go works by looking a fixed distance ahead and reading a value off the horizon, on the assumption that both sides play well from there. That assumption fails completely under hidden information, and Brown and Sandholm make the failure vivid with a one-shot sequential Rock–Paper–Scissors: if the searcher assumes the opponent will play the equilibrium (each throw one-third of the time) beyond the horizon, then every action looks equally valued at zero, so the searcher might settle on “always play Rock” — whereupon the opponent switches to always Paper and the true value plummets. A leaf in an imperfect-information game therefore has *no single value*: its worth depends on the strategy the searcher will adopt, which is precisely what the search is trying to determine.

Pluribus’s central contribution is the way it gives leaves a value anyway. When search reaches the depth limit, instead of freezing one continuation, **each player still in the hand chooses among four different continuation strategies** for the remainder of the game, and may mix over them. The four are deliberately simple: the blueprint itself, and three biased copies that multiply the probability of *folding*, of *calling*, and of *raising* (each then renormalized). Because an opponent can always switch to whichever continuation punishes the searcher most, an unbalanced strategy — the poker equivalent of always playing Rock — is no longer rewarded, and the searcher is driven toward balance. This idea was first proven in a two-player precursor, Modicum (Brown, Sandholm & Amos, 2018), which beat two former champion bots while running on a 4-core laptop with 16 GB of memory — a striking sign that depth-limited search with continuation strategies could stand in for a supercomputer.

Pluribus generalizes the idea from two players to six and adds a subtle but important twist. In Modicum only the *opponent* chose among continuation strategies while the searcher always played the blueprint — sound in two-player zero-sum, but it hands the opponent the initiative and leaves the searcher timid and low-value. Pluribus lets the **searcher choose among the continuation strategies too**, which balances the players and, in the authors’ words, is “more effective, easier, and more elegant.” It also re-solves not from the current decision point but from the *start of the current betting round*, holding fixed only the actions it has already taken. This “unsafe” search — so named because, unlike Libratus’s, it carries no exploitability guarantee — is cheaper (most six-handed hands are folded immediately, so few need a strategy at all) and, because it begins just after a high-branching chance event, turns out to be hard to exploit in practice.

What is conspicuously missing from all of this is a *guarantee*. DeepStack bounded its exploitability by $k_1\epsilon + k_2/\sqrt{T}$ and Libratus by 2Δ ; Pluribus offers no such bound, and the omission is principled rather than careless. CFR’s engine still does, in any finite game, drive each player’s *average regret* to zero,

$$\frac{R_i^T}{T} \rightarrow 0 \quad (\text{no-regret}),$$

but the inference that carried the two predecessors from no-regret to safety — *no-regret play converges to a Nash equilibrium, and a Nash equilibrium cannot be beaten* — holds only when there are two players and the game is zero-sum. With six players the first implication fails (self-play need not approach an equilibrium) and the second is meaningless (an equilibrium is not unbeatable). The very iterations that *proved* DeepStack and Libratus safe therefore buy Pluribus only empirical strength: superhuman in practice, with nothing certified. To make those iterations cheap enough to run six-handed, Pluribus leans on **Linear CFR** — weighting iteration t ’s contribution by t so the bad early iterations wash out about three times faster — and a **modified negative-regret pruning** that skips hopeless actions on most iterations (everywhere but the final round); together with lazy memory allocation, these are what shrink the bill to a single server.

6.4.4 Caveats, dead-ends, and what the paper under-describes

As with both predecessors, the published article is the clean story and the engineering honesty lives in the supplement — only more so here, because Pluribus is a six-page *Science* paper whose architecture is almost entirely relegated to its supplementary materials and to two companion papers. The most consequential admission is that Pluribus uses **unsafe** subgame solving — it assumes opponents have played the strategy it computes *for* them — which, the authors state plainly, “lacks theoretical guarantees on performance even in two-player zero-sum games and there are cases where it leads to highly exploitable strategies.” Safe alternatives exist, but in head-to-head play they did worse, so Pluribus takes the empirical win and mitigates the risk only by always re-solving from the start of the betting round. A second seam is inherited and never fully closed: on the *first* betting round, opponent bets too far off the blueprint’s menu are still **rounded** by action translation — the very weakness Libratus’s self-improver existed to chip at — and Pluribus, having no self-improver, simply lives with it.

Several other asterisks matter. Pluribus’s headline innovations are **never individually ablated**: the authors concede that the variance of no-limit poker and the cost of human trials make it “too expensive” to measure each one’s contribution, so the reported component speedups (roughly $3\times$ from Linear CFR, $2\times$ from the modified pruning, more than $2\times$ in memory from lazy allocation) are estimates, and the overall winning margin is not decomposed. The supplement does, however, dispatch one tempting misconception: assuming a *single* blueprint continuation at the leaves — the obvious thing to try, and the way the underlying study notes sometimes gloss the method — was shown in the two-player precursor to *lose* to both champion bots (by 10 and 1 mbb/g), whereas the four-continuation version *won* (by 6 and 11); the continuation-strategy set is doing real work, not decoration. Smaller curiosities round out the picture: Pluribus plays its **final** search iterate rather than the usual time-average, to avoid residual bad actions; it learned to **abandon “limping”** during self-play yet **“donk-bets” far more than humans do**; and — like Libratus — its **code was never released**, leaving only pseudocode for independent verification, because poker is played commercially.

6.4.5 Compute & accessibility

Pluribus’s compute story is the one most people remember, and it genuinely inverts its predecessors’. The blueprint was trained in **eight days on a single 64-core server** for about **12,400 core-hours** and under 512 GB of memory — roughly **\$144** at cloud spot prices — and at the table Pluribus runs on **two CPUs (28 cores) and under 128 GB, with no GPUs at any point**, taking one to thirty-three seconds per decision and playing about twice as fast as a human. Set against the field the contrast is almost comic: AlphaGo used 1,920 CPUs and 280 GPUs, Deep Blue 480 custom chips, and Libratus around fifteen million core-hours to build its blueprint and a roughly hundred-CPU cluster to play. Pluribus reached a *harder* milestone — more players, a larger game — for on the order of a thousandth of Libratus’s training compute. Where DeepStack concentrated its cost offline and Libratus paid heavily at both ends, Pluribus is **cheap at both ends**, and that, as much as the six-player result, is the paper’s thesis.

This reframes accessibility entirely. For the first time a superhuman poker system was reproducible, in principle, by a single well-equipped researcher rather than a supercomputing centre — the authors explicitly present it as a rebuttal to the worry that frontier game-AI would belong only to teams with millions of dollars of hardware. The collapse is not magic but algorithmic: the compounding of depth-limited search (which the authors estimate saves at least five orders of magnitude over solving to the end), Linear CFR, and aggressive pruning and memory thrift. The one caveat is that the artefact itself stays closed — no code — so “accessible” describes the *method*, demonstrated at laptop scale by Modicum, more than a downloadable program.

6.4.6 Strengths and limitations

Pluribus’s signal strength is simply that it is **first**: the first AI to reach superhuman performance in any widely recognized benchmark game with more than two players or two teams, and in poker’s most popular form. The win was decisive and rigorous — **+48 mbb/g ($p = 0.028$)** against five elite pros at the table, and **+32 mbb/g ($p = 0.014$)** with five copies against a lone pro — measured with the

AIVAT variance reducer (which cut variance about ninefold) over tens of thousands of hands, against thirteen professionals who had each won over a million dollars and who had days to hunt for weaknesses; the win rate barely wavered. A later rematch even beat **Linus Loeliger**, widely regarded as the best six-max cash player alive. And, like Libratus, Pluribus is **robustness-first**: it plays a fixed strategy, never models or adapts to opponents, and does not even know their identities, so it cannot be lured into a counter-exploitable adjustment — a discipline that, with **no human data and no domain knowledge**, keeps the result clean.

The limitations are precisely what it gave up to get there. Pluribus has **no safety guarantee whatsoever** in the six-player setting — no Nash convergence, no exploitability bound — so its superhuman status is an *empirical* fact about thirteen strong humans over tens of thousands of hands, not a theorem; a sufficiently coordinated table, or simply a different game, carries no assurance. It leans on **unsafe** search, still **rounds off-tree bets on the first round**, and remains **purely tabular and abstraction-based**, with nothing learned that generalizes across situations — the blueprint is a giant lookup table, not a model. The authors add that the whole approach may not survive where players can **communicate and collude**, which poker largely forbids. And its 48-mbb/g six-handed win rate, though decisive, is not the same currency as Libratus’s 147 heads-up — different game, five opponents, higher variance. Most of these gaps are picked up by later systems; the one this thesis singles out — that **multiplayer success came with no safety guarantee at all** — is the explicit target of Contribution 2.

6.4.7 Legacy and modern relevance

Two ideas outlast the poker specifics. The first is technical: **depth-limited search made sound under hidden information by giving each leaf a small menu of selectable continuation strategies rather than a single value** — a clean way to look only a little way ahead in a game where, naïvely, you cannot. The second is economic, and the field absorbed it most deeply: Pluribus is the canonical demonstration that **a harder problem can be solved with a thousandfold less compute through better algorithms and search at decision time**, rather than more hardware. Noam Brown has since pointed to exactly this — that adding real-time search was worth as much as an enormous scaling of the precomputed strategy — as an early, concrete instance of the “**test-time compute**” argument now central to reasoning models such as o1. In the architectural arc of this chapter, Pluribus is the point where the *player-count* barrier falls; the *generalization* barrier it leaves standing — everything is still tabular and hand-abstracted — is what **ReBeL** and **Student of Games** dismantle next, by replacing the blueprint with learned belief-state values.

Several of Pluribus’s parts remain directly reusable: the **continuation-strategy (multi-valued leaf)** construction for robust depth-limited search in any hidden-information or multi-agent setting; **Linear (discounted) CFR**, now a standard convergence accelerator and the launch point for later discounted-CFR variants; **regret-based pruning** for scaling CFR; **AIVAT** as a low-variance evaluator; and the *add-the-action-and-re-solve* handling of off-tree bets inherited from Libratus. What is superseded is the surrounding tabular edifice — hand-tuned abstraction and a lookup-table blueprint — which the learned representations of ReBeL and SoG dissolve, along with the *unsafe* search those belief-state methods make

unnecessary. The honest verdict mirrors Libratus’s: **superseded as a deployed architecture, but vindicated and sharpened as a thesis**. For this dissertation specifically, Pluribus is the keystone of Contribution 2 — the empirical proof that Nash-and-search methods *work* in N-player imperfect-information games while offering **no safety guarantee at all**, which is exactly the gap a theory of multi-agent safe exploitation must close.

6.5 ReBeL (2020)

Pluribus broke the player-count barrier, but it did so while remaining exactly what Libratus was — a giant, hand-abstracted lookup table with nothing learned that transfers from one situation to the next — and it bought its six-handed win by giving up safety altogether, leaning on *unsafe* search with no exploitability guarantee of any kind. ReBeL (Brown, Bakhtin, Lerer & Gong, 2020; Facebook AI Research) steps back from six players to two and asks the opposite question: **what if, instead of hand-crafting an abstraction and precomputing a blueprint, we ran AlphaZero — self-play reinforcement learning plus search, at both training and test time — in a game of hidden information, recovering the very guarantees Pluribus discarded while throwing the abstraction out entirely?** Its answer, *Recursive Belief-based Learning*, is the first algorithm to make reinforcement learning *and* search provably sound in imperfect-information games. The trick is to recast the game as a “perfect-information” game over **public belief states** — probability distributions over what each player might be holding, given common knowledge — on which value and policy functions are well-defined; an AlphaZero-style loop then trains a neural value (and policy) network on those states, with counterfactual regret minimization (CFR) solving depth-limited subgames at the leaves. ReBeL provably converges to a Nash equilibrium in any two-player zero-sum game, beat the top human heads-up specialist Dong Kim by 165 mbb/g (milli-big-blinds per game, the win-rate unit from the previous sections) over 7,500 hands while using *far less* domain knowledge than any prior poker AI — and, unlike the closed Libratus and Pluribus, its implementation (for Liar’s Dice) was **open-sourced**.

At a glance	ReBeL (2020)
Players	2 (heads-up) — <i>guarantees</i> are two-player zero-sum; the algorithm generalizes to more players but without the guarantees
Game type	General 2p0s imperfect-information; evaluated on HUNL poker + Liar’s Dice (and turn endgame hold’em). Reduces to an AlphaZero-like algorithm in perfect-information games
Blueprint (offline)?	No stored blueprint — the offline product is a <i>learned PBS value (+ policy) network</i> from self-play, not a strategy table

At a glance	ReBeL (2020)
Neural component	PBS value network + (optional) PBS policy network; MLP (GeLU/LayerNorm), 6×1536 hidden for poker, input = belief over each player's 1,326 hands + board + pot + bet flag
Search mechanism	CFR (CFR-D / CFR-AVG; also FP/FLOP) solving a depth-limited subgame rooted at a PBS, with the learned value net supplying (iteration-dependent) leaf values — at both training and test time
Abstraction?	None — no card/information abstraction (lossy or lossless); the value net replaces it. Keeps only a small (9) hand-chosen bet-size menu, with off-tree bets added live
Perfect-info too?	No (presented/evaluated as imperfect-information), with the nuance that it <i>degenerates</i> to AlphaZero-style search if private information is removed — full unification is Student of Games' claim
Compute	GPU-trained: full HUNL used ~ 90 DGX-1 nodes $\times$ 8 V100 GPUs for self-play data generation (a contrast with Pluribus's CPU-only $\sim \$150$); CFR on a single CPU thread; play < 2 s/hand, ~ 5 s/decision
Key innovation	Public belief states + an AlphaZero-style self-play loop training a PBS value/policy net with CFR run in belief space: the first <i>sound</i> RL+Search for imperfect-information games, recovering a provable 2p0s Nash guarantee with no abstraction or blueprint

6.5.1 The gap it closed

The previous three systems converged on one recipe — a precomputed strategy plus real-time search — and Pluribus pushed it to six players, but every one of them was, underneath, a *tabular* object: a strategy or blueprint stored over a hand-engineered abstraction, with nothing learned that generalizes to a situation never explicitly solved. DeepStack alone had a neural component, yet even it leaned on hand-crafted features, a 1,000-bucket clustering at the network's input, and an abstraction on the river. Meanwhile the most successful paradigm in all of game AI — AlphaZero's marriage of self-play reinforcement learning with search, which learns its own evaluation from scratch and reuses it both to train and to play — had

been *unavailable* for imperfect information. The open question ReBeL answers is the one Noam Brown calls the field’s “holy grail”: **can the AlphaZero recipe be made to work, soundly, in games of hidden information?**

The reason it could not, before, is subtle and is the conceptual crux of the whole chapter. AlphaZero assumes each state has a single well-defined value: a chess position is worth what it is worth, regardless of how often you reach it. Hidden information destroys that assumption. The paper makes it vivid with a modified Rock–Paper–Scissors in which scissors wins (or loses) double: the equilibrium is to throw rock and paper 40% of the time and scissors 20%, and at that equilibrium every action has expected value zero. If Player 1 runs the one-ply lookahead search that works in chess — substituting the equilibrium value at the leaves — every move looks equally good, so the searcher might settle on “always rock”, whereupon the opponent switches to “always paper” and rock’s true value collapses from 0 to -1 . **In an imperfect-information game the value of an action depends on the probability with which it is played**, so a state defined by the sequence of actions alone has no unique value — and AlphaZero-style search is simply unsound. Pluribus had patched the symptom with selectable continuation strategies at the leaves; ReBeL cures the disease by changing what a “state” *is*. It redefines the state to include the probability distribution over the hidden information — a *public belief state* — on which, as the next sections show, values become well-defined again and the entire AlphaZero machinery can be ported across. In doing so it also dissolves the two crutches DeepStack and the abstraction systems still leaned on: the hand-crafted abstraction and the precomputed blueprint both disappear, replaced by a value function learned purely from self-play.

6.5.2 Architecture

ReBeL is best read as **AlphaZero for imperfect information**: a self-play loop that trains neural value and policy networks, where the “search” used during both training and play is CFR solving a depth-limited subgame — but everything operates on *public belief states* rather than raw game states, and the leaf evaluator is a learned value network rather than a rollout (Figure 6.4).

The loop has three moving parts. **Self-play reinforcement learning** drives the whole thing: starting from a root PBS, ReBeL constructs a depth-limited subgame, solves it, records the solved values and policy as training targets, samples a leaf to become the next root, and continues to the end of the game — then retraining the networks and iterates, exactly as AlphaZero alternates self-play with network training. **Search** is the inner solve: ReBeL runs CFR (specifically CFR-D, “CFR with decomposition”, or its variant CFR-AVG) over the depth-limited subgame, and on every CFR iteration it sets each leaf’s value by querying the **learned PBS value network** — so the leaf values shift from iteration to iteration, which is precisely what keeps the search sound under hidden information. The **neural networks** are the learned intuition: a value network that maps a PBS to the values of the player’s possible hands, and an optional policy network that warm-starts CFR to cut the number of iterations needed.

The architectural signatures are all *absences* relative to the earlier systems. There is **no precomputed blueprint** — the only artefacts carried out of training are the two networks. There is **no abstraction**

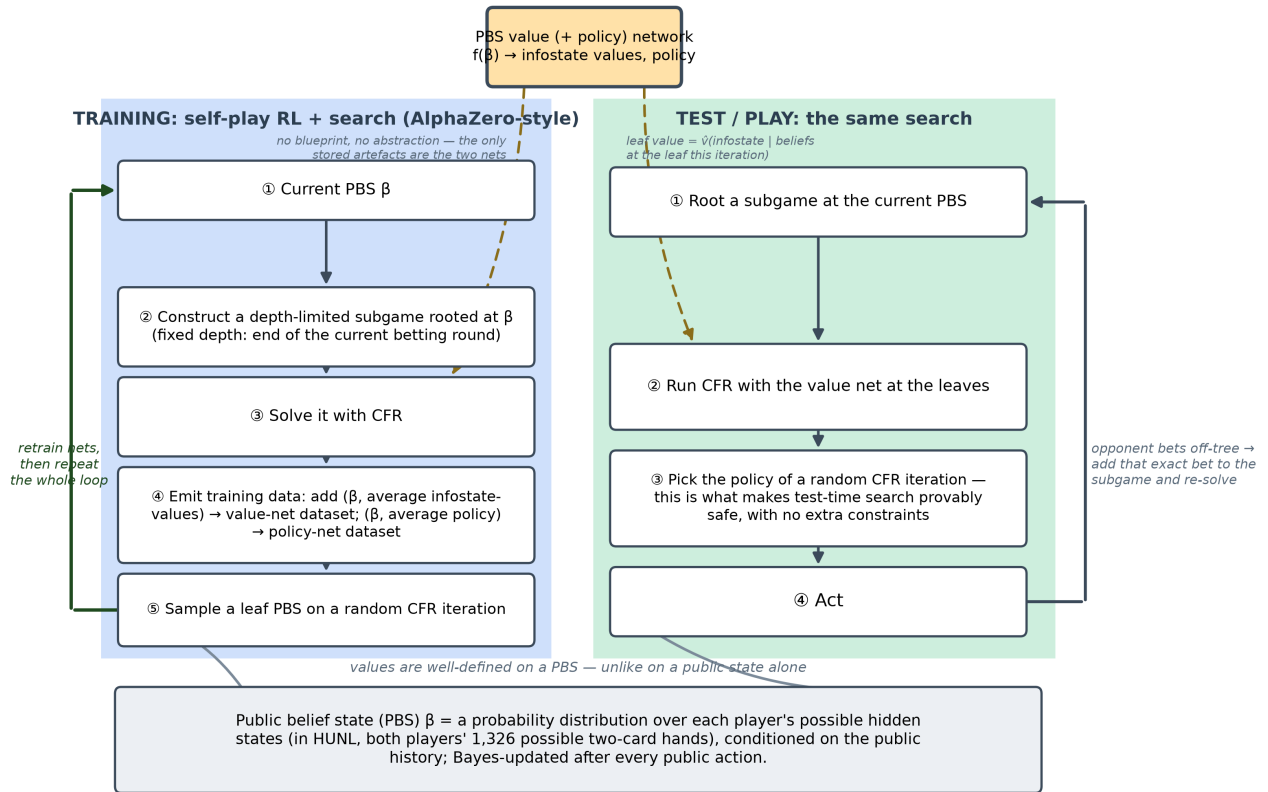


Figure 4: ReBeL’s AlphaZero-style loop: self-play training (left) and test-time play (right) share one PBS value/policy network, both solving depth-limited subgames rooted at the public belief state (shared definition box, bottom).

of any kind: where DeepStack, Libratus, and Pluribus all bucketed strategically similar hands, ReBeL computes a distinct policy for every infostate, feeding the network only the raw belief distribution over both players’ 1,326 possible hands plus the board, pot, and a single “has anyone bet this round” flag. And the CFR search runs on a **single CPU thread** with no abstraction at all; the heavy compute is the GPU self-play that trains the value net (a balance discussed under *Compute*). The classic building blocks of the earlier steps are still visible — CFR (Step 3) is the search engine and neural value approximation (Step 5) is the leaf evaluator — but they are now fused into a single self-play-plus-search loop, with the public belief state (the section below) as the representation that makes the fusion possible.

6.5.3 Key innovation: public belief states and learned values in belief space

ReBeL’s contribution is one conceptual move with three technical consequences. The move is to stop searching over *states* and start searching over *beliefs about states*.

The **public belief state (PBS)** is the heart of it. The Meta AI blog gives the cleanest intuition. Take a card game and modify it so the players cannot see their own cards — only an impartial referee can; on each turn a player announces, for every card they *might* hold, the probability with which they would take each action, and the referee samples the move for the player’s true card. Because all players’ strategies are assumed common knowledge, everyone can track, via Bayes’ rule, the probability that each player holds each possible hand. This modified game is **strategically identical** to the original, yet it contains *no private information*: its state — the vector of those probabilities — is fully observed by everyone. ReBeL calls that state a public belief state: formally, a joint probability distribution over the players’ possible infostates, given the common public observations. Crucially, viewing imperfect-information games as continuous-state perfect-information games this way is an old idea (it goes back to work on cooperative POMDPs); ReBeL’s achievement is being the first to combine it with self-play reinforcement learning in an *adversarial* setting.

The first consequence is that **values become well-defined again**. An imperfect-information subgame rooted at a mere *public state* has no unique value, because its worth depends on the hidden hands that reach it — which is exactly why heuristic search failed. But a subgame rooted at a *public belief state* does have one: in two-player zero-sum games every PBS β carries a unique value $V_i(\beta)$ with $V_1(\beta) = -V_2(\beta)$, defined by both players playing a Nash equilibrium in the subgame from β . This is precisely the property AlphaZero relies on in chess, now recovered for poker — and it *formalizes and generalizes DeepStack’s belief-conditional counterfactual values*, which were the first instance of a PBS value function but were trained in a much more ad-hoc way.

The second consequence is that the AlphaZero loop can run, with **CFR as the search algorithm**. One could in principle run AlphaZero’s Monte-Carlo tree search directly on the belief game, but the belief representation is a very high-dimensional continuous space (in the blog’s small card example the action space alone has 156 dimensions), and tree search is hopeless there. The saving grace is that in two-player zero-sum games these belief representations are *convex* optimization problems, and CFR is effectively a gradient-based solver for them. So ReBeL searches with CFR rather than MCTS — the one substantive

difference from AlphaZero. A technical subtlety the paper proves (Theorem 1) is that the quantities CFR actually needs, the per-infostate values, are a *supergradient* of the concave PBS value function, so ReBeL learns an infostate-value network rather than a single scalar per PBS.

The third consequence — and the headline — is that the whole thing is **provably sound**, recovering the guarantee Pluribus surrendered. The deepest result concerns *test time*. During self-play ReBeL can assume both players’ policies are common knowledge, so it always knows the true PBS; but against a real opponent it does not know their policy and therefore does not know which PBS it is in, which naively breaks search. ReBeL’s elegant fix is to run *the same algorithm* at test time and **act according to the policy of a randomly chosen CFR iteration**. The paper proves (Theorem 3) that this yields safe search — a Nash equilibrium *in expectation* — with no extra constraints, where every prior “safe search” method bolted on constraints so costly they were never fully used in a competitive bot. Concretely, with a PBS value network whose error is at most δ and T CFR iterations per subgame, ReBeL plays an ε -Nash equilibrium — one no opponent can exploit for more than ε — with

$$\varepsilon = \delta C_1 + \frac{\delta C_2}{\sqrt{T}},$$

for game-specific constants C_1, C_2 . The structure deliberately echoes DeepStack’s $k_1\epsilon + k_2/\sqrt{T}$ — a value-error term plus an ordinary CFR-convergence term — but where DeepStack’s bound governed the exploitability of a single re-solve, ReBeL’s governs *convergence to a Nash equilibrium of the whole game*, and both terms now scale with the learned value’s error δ : as the value network improves, $\delta \rightarrow 0$ and ReBeL’s play approaches an exact Nash equilibrium. This is the precise sense in which ReBeL *returns to two-player zero-sum and recovers the theoretical guarantees Pluribus gave up* — the same $1/\sqrt{T}$ CFR backbone that proved DeepStack and Libratus safe, now wrapped around a self-play-learned belief-state value instead of a hand-built abstraction.

6.5.4 Caveats, dead-ends, and what the paper under-describes

ReBeL inverts the usual pattern of this chapter: the *theory* is in the main text and the *engineering honesty* — especially the precise sense in which it uses “far less domain knowledge” — is relegated to the appendices.

The most important detail is what ReBeL deliberately **throws away**, catalogued in Appendix D. Every prior top poker AI, DeepStack included, used **information abstraction** to bucket strategically similar hands; ReBeL uses *none*, lossy or lossless, computing a unique policy per infostate from the raw belief distribution. DeepStack trained its value net on *randomly generated* PBSs drawn from a hand-tuned sampler; ReBeL generates its training PBSs purely from self-play, arguing that random sampling “would be like learning a value function for Go by randomly placing stones on the board” — and indeed it shows (Figure 2) that a value net trained on random beliefs “fails to learn anything valuable.” Prior agents precomputed exact all-in equity tables and solved *to the end of the game* on the third betting round (using river abstraction to make that tractable); ReBeL does neither, always solving only to the end

of the *current* round and learning even the all-in values itself. The upshot is honest but double-edged: ReBeL must learn *six* “layers” of values where DeepStack needed three, which *increases* the surface for error propagation — a real cost accepted in exchange for shedding the abstraction. And it is not domain-knowledge-free: it keeps a hand-chosen menu of at most eight or nine bet sizes (perturbed during training), the one concession to tractability, with off-tree bets handled by adding them to the subgame and re-solving, à la Libratus and Pluribus.

Several other asterisks matter. The clean theorems rest on **idealizations**: Theorem 2’s convergence assumes a perfect function approximator, and the guarantees assume players’ policies are common knowledge (the §6 result removes the *test-time* version of that assumption, but the training analysis still idealizes the network). The variant the agent actually runs for efficiency, a modified **CFR-AVG**, is by the authors’ own admission *not known to be theoretically sound* — “whether or not this modified form of CFR-AVG is theoretically sound remains an open question” — even though it performs well in poker, a familiar gap between the proven algorithm and the shipped one. The safe-search guarantee also depends on picking a *random* CFR iteration, which could land on a terrible early one; this is mitigated only by using Linear CFR, which down-weights early iterations. And as with its predecessors, essentially all of the architecture, hyperparameters, and the all-important compute figures live in the appendices rather than the six pages of main text — the paper is, by design, a theory paper.

6.5.5 Compute & accessibility

ReBeL’s cost profile is the mirror image of Pluribus’s, and the contrast is the sharpest in the chapter. Where Pluribus’s intuition was a blueprint solved by **CPU-only CFR on a single 64-core server for about \$150**, ReBeL’s intuition is a neural network, and training it is **GPU-cluster-scale**. The bottleneck is generating self-play data — sequential CFR solves whose every iteration evaluates all leaf nodes through the network — so the paper parallelizes data generation across “up to 128 machines with 8 GPUs each”, and the full HUNL agent in particular used **about 90 DGX-1 nodes, each with eight 32 GB Nvidia V100 GPUs** (on the order of 700 GPUs), feeding a single training machine over 1,750 epochs of millions of examples apiece. This is squarely a deep-learning training bill, not a CFR bill: the CFR search itself runs on a single CPU thread. At the table, though, ReBeL is cheap and DeepStack-like — under two seconds per hand on average and never more than five seconds per decision, with preflop subgames cached to go faster still.

The accessibility story splits in two, and it is the cleanest contrast with the earlier poker systems. On the one hand ReBeL’s *method* is the most general and least hand-tuned of the four, and — decisively — its **implementation was open-sourced** (for Liar’s Dice), where the code for both Libratus and Pluribus was kept closed. That open release is a genuine step toward reproducibility that the CMU systems never offered. On the other hand, the headline *poker* result was *not* released — the authors withheld it on the explicit grounds that ReBeL “can compute a policy for arbitrary stack sizes and arbitrary bet sizes in seconds”, making it a ready-made cheating tool — and reproducing that result at full strength required a large V100 fleet most groups did not have. So ReBeL is “open” in a way its predecessors were not, but the open artefact is the research game, not the superhuman poker bot.

6.5.6 Strengths and limitations

ReBeL’s signal strength is **generality with a guarantee**. It is the first algorithm to make the AlphaZero RL+Search paradigm *sound* in imperfect-information games, it provably converges to a Nash equilibrium in any two-player zero-sum game, and it backs the theory with results: it beat the prior champions Slumbot (+45 mbb/g) and BabyTartanian8 (+9 mbb/g), drove the local-best-response probe to a large loss, and beat the top human specialist Dong Kim by 165 mbb/g over 7,500 hands — all while using *far less domain knowledge than any prior poker AI*, with no card abstraction and the most general representation of the four. The same algorithm, unchanged, converges to approximate Nash in Liar’s Dice, demonstrating that this is a *framework* and not a poker program; in perfect-information games it gracefully reduces to an AlphaZero-like method. And it recovers, for the neural era, the safety that Pluribus had to abandon — search that is provably safe at test time with no bolt-on constraints.

The limitations are precise and they set up the final system. ReBeL’s guarantees are **two-player zero-sum only**: the unique PBS value, the convexity that licenses CFR-as-search, and the soundness proofs all lean on that structure, and although the algorithm has been run with more players, the theory does not follow it there — the scorecard’s “2 players” is a statement about the *guarantee*, not the code. Its most concrete scaling wall is that the **PBS grows with the amount of hidden information**: the network’s input scales with the number of infostates in a public state, so games with great strategic depth but little common knowledge — the paper names Recon Blind Chess — blow the representation up, and adding players only makes the belief space larger. It assumes the **exact rules of the game are known** (a MuZero-style extension to unknown dynamics is flagged as future work), it inherits a residual **value-approximation error** δ that bounds how close to Nash it actually plays, and its strongest empirical result rests on a **single human opponent** over 7,500 hands rather than a multi-pro field.

6.5.7 Legacy and modern relevance

Strip away the poker and ReBeL’s enduring idea is a representational one: **convert hidden information into a belief state and the full power of self-play reinforcement learning plus search transfers across**. Public belief states are now the standard substrate for sound search in imperfect-information games, and ReBeL is the cleanest statement of “AlphaZero for imperfect information” — search woven into *both* training and play. Its direct descendant is **Student of Games** (2023), which carries the belief-state-plus-sound-search idea further into a single algorithm spanning perfect *and* imperfect information; the same RL-plus-search-with-learned-models lineage also runs into **CICERO** (2022), Meta’s human-level Diplomacy agent, which plans with learned models in a seven-player, mixed-motive, natural-language game far outside ReBeL’s two-player guarantees. More broadly, ReBeL belongs to the line of work — DeepStack, Libratus, Pluribus, and now ReBeL — that Noam Brown points to as an early, concrete instance of **test-time compute**: spending computation on search at decision time, guided by learned intuition, rather than baking everything into one precomputed policy, the same thesis now central to reasoning models.

Several subsystems remain directly reusable. The **PBS representation** itself is the most important

— a belief distribution over opponents’ hidden states given common knowledge — and it is the natural seed for belief-based opponent modelling: this thesis’s Contribution 1 extends PBS to carry beliefs about the opponent’s *strategy type*, not just their cards. The **self-play-trained value/policy function on belief states**, the pattern of **CFR-as-search with a learned leaf evaluator**, the **random-iteration “safe search for free”** trick, and the simple fast equilibrium finder **FLOP** the paper introduces are all transferable. What is superseded is mostly DeepStack’s hand-engineered scaffolding, which ReBeL’s no-abstraction self-play dissolves, and then ReBeL itself as the *general* framework, which Student of Games subsumes. The honest verdict is that **ReBeL is a living stepping stone, not an endpoint**: as a deployed poker artefact it was never released and is now eclipsed, but as an idea — public belief states plus AlphaZero-style RL and search — it is foundational and very much alive. For this thesis specifically it cuts two ways at once: it is the starting point for Contribution 1’s richer belief state, and it is the proof that the soundness Pluribus abandoned *can* be recovered — just not yet beyond two players, which is exactly the frontier Contribution 2 sets out to cross.

6.6 Student of Games / SoG (2023)

ReBeL recovered soundness for two-player zero-sum poker and discarded both the abstraction and the blueprint — but it remained an imperfect-information method that merely *reduced* to AlphaZero in perfect-information games rather than unifying the two, solved a *fixed* depth-limited subgame, and tied its test-time search to its training procedure. Student of Games (Schmid et al., 2023; Google DeepMind, with collaborators at the University of Alberta and EquiLibre Technologies in Prague) closes the last gap in the chapter by asking the most ambitious question of all: **what if a single algorithm, learning from self-play with no human data, could play chess, Go, heads-up poker, and Scotland Yard — growing its own search tree as needed and remaining provably sound for perfect- and imperfect-information games alike?** Its answer fuses the chapter’s two great lineages — it is, in first author Martin Schmid’s words, “AlphaZero and DeepStack in a single big unified algorithm.” The engine is **Growing-Tree counterfactual regret minimization (GT-CFR)**: an anytime search that builds the game tree incrementally, guided by a policy network, with a learned value network evaluating the leaves it has not yet expanded. On perfect-information subtrees GT-CFR behaves like AlphaZero’s MCTS; on imperfect-information ones it behaves like CFR; and a single soundness theorem covers both. The system reaches **strong amateur-to-professional play in chess and Go, beats Slumbot — the strongest openly available heads-up no-limit bot — and defeats the state-of-the-art Scotland Yard agent**, all while being **proven to converge toward minimax-optimal play as computation and network accuracy grow**. It shares several DeepStack authors (Bowling, Moravčík, Burch, Schmid) — the Alberta lineage that began this chapter, here in contrast to ReBeL’s Facebook/FAIR provenance — and it first appeared in 2021 under the name *Player of Games* before being renamed for its 2023 *Science Advances* publication.

At a glance	Student of Games (2023)
Players	2 (two-player zero-sum) — guarantees <i>and</i> evaluation are 2p0s (Scotland Yard’s detectives count as one team); the underlying formalism is more general, but Nash is “less meaningful” beyond 2p0s
Game type	Both perfect- and imperfect-information — the only unified system: chess + Go (perfect) and HUNL poker + Scotland Yard (imperfect)
Blueprint (offline)?	No — the offline product is a learned value+policy network, not a stored strategy table (as in ReBeL)
Neural component	A single counterfactual value-and-policy network (CVPN) : one net outputs per-infostate counterfactual <i>values</i> and a prior <i>policy</i> , for all game stages
Search mechanism	GT-CFR — alternates a CFR regret-update phase (CVPN values at the leaves) with an AlphaZero-style PUCT expansion phase that <i>grows</i> the public tree; run inside continual re-solving, at both training and test time
Abstraction?	None of the card/information kind (the CVPN replaces it); keeps only a small <i>randomized betting</i> (action) menu in poker (20,000 → 4–5 actions), like ReBeL
Perfect-info too?	Yes — the distinguishing row; the only system in the chapter demonstrated on perfect-information games, with the <i>same</i> algorithm and soundness covering both classes
Compute	TPU-trained, deliberately matched to AlphaZero’s budget (Google TPuv4; Go the most expensive); reported <i>relative</i> to AlphaZero, no single dollar figure; search is $O(kT^2)$ ($O(T)$ for perfect-info); the full agent/code was not released
Key innovation	Growing-Tree CFR + sound self-play: one self-play-with-search algorithm, sound for both game classes , that grows its tree incrementally with a CVPN at the leaves and provably converges to Nash

6.6.1 The gap it closed

For seventy years, the two great traditions of game AI ran on separate tracks. One — minimax, alpha-beta, Monte-Carlo tree search, and ultimately AlphaZero — conquered *perfect-information* games by combining search with a learned value function, and its single most general expression, AlphaZero, mastered chess, shogi, and Go with one algorithm. The other — counterfactual regret minimization and the four poker systems of this chapter — conquered *imperfect-information* poker through game-theoretic reasoning. But the tracks never met. AlphaZero could not play poker: its Monte-Carlo tree search is *unsound* under hidden information, for the reason the ReBeL section made precise — the value of a state depends on the probability with which you reach it, so values cannot simply be backed up from local subtrees. And the poker systems could not play Go: they were built around the betting structure of one game. Even ReBeL, the most general of them, was presented and guaranteed only for imperfect information, *reducing* to AlphaZero in the perfect-information case rather than genuinely spanning both.

The open question Student of Games answers is whether a *single* algorithm — one search procedure, one network architecture, one self-play loop — can be **sound and strong across both game classes at once**. The difficulty is not merely engineering. A perfect-information search wants to expand the one best line deeply and read a single value off each leaf; an imperfect-information search must keep a *distribution* over actions (so as not to leak private information) and must reason about a *belief* over hidden states at every leaf. Reconciling these in one algorithm means finding a search that grows a tree the way AlphaZero does — guided, asymmetric, anytime — while computing the *game-theoretically sound* quantities that CFR does. Student of Games closes the gap by building exactly that search, GT-CFR, on top of the public-belief- state representation ReBeL had established, and by training its networks through a *sound self-play* loop whose targets come from the search itself. The result is the chapter’s capstone: the point where the abstraction-to-neural and offline-to-search arcs are joined by a third — perfect and imperfect information, unified.

6.6.2 Architecture

Student of Games is, structurally, **AlphaZero’s self-play loop with Monte-Carlo tree search replaced by GT-CFR and with the value/policy network defined over public belief states** — an offline phase that learns a network by self-play and an online phase that searches with it, but where the same search now runs in both phases (Figure 6.5).

The representation is inherited from the ReBeL lineage: a **public belief state** $\beta = (s_{\text{pub}}, r)$ pairs the public state (in poker, the betting history and board) with a **range** r — a pair of distributions over the information states each player could privately occupy. Perfect-information games are simply the degenerate case in which every public state has exactly *one* information state and the belief is a point mass, which is precisely why one representation can serve both classes. On this representation sit the two components. The **counterfactual value-and-policy network (CVPN)** is a *single* network that, given a belief state, emits both a vector of counterfactual values (one per information state, per player) and a prior policy — one network doing both jobs, for every stage of the game, where DeepStack used

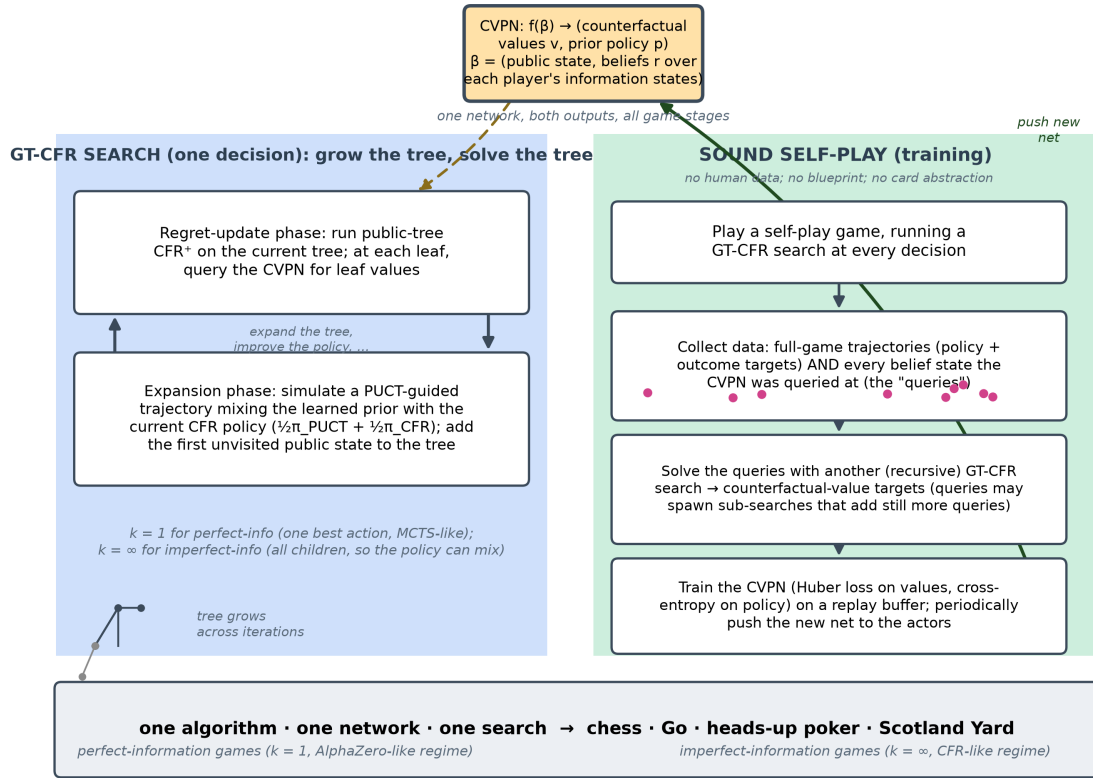


Figure 5: Student of Games: the GT-CFR search loop (left) and the sound self-play training loop that feeds it (right) share one CVPN; the footer strip names the unification across perfect- and imperfect-information games.

separate value-only networks per round. The **GT-CFR search** is the engine that turns that network into a strategy, and the **sound self-play** loop is what trains the network from the search’s own output. Offline, actors play self-play games, run a GT-CFR search at every decision, and emit two kinds of training data — full-game trajectories and the belief states the network was queried at during search — while trainers fit a new CVPN and periodically push it back. Online, the agent runs the very same GT-CFR search (via continual re-solving) to choose each move. The classic building blocks are all visible — CFR (Step 3) is the search’s inner loop, value-and-policy approximation (Step 5) is the CVPN, and public belief states and decomposition (the DeepStack/ReBeL lineage) are the representation — but the binding novelty is GT-CFR, the search that *grows* its tree, and the sound self-play that keeps every search consistent with every other.

6.6.3 Key innovation: Growing-Tree CFR and sound self-play

Student of Games’ contribution is a search algorithm that does for imperfect information what Monte-Carlo tree search did for perfect information — grow a tree intelligently toward the lines that matter — *without* sacrificing the soundness that hidden information demands. Two ideas make it work, and a single theorem makes it general.

The first idea is **growing the tree by alternating two phases**. AlphaZero’s MCTS expands one node per simulation and never looks back, because in a perfect-information game the value of a node, once computed, never changes — the future does not alter the past. Under imperfect information this fails: as Schmid puts it, observing an opponent’s *future* action changes your *belief* about the private state they held in the past, so changing the strategy anywhere in the tree ripples everywhere. A search therefore cannot just expand-and-evaluate; after each expansion it must re-solve the *whole* tree to keep it self-consistent. GT-CFR does exactly this. Each iteration runs two phases in turn. The **regret-update phase** runs several iterations of public-tree CFR over the *current* tree, and wherever the tree stops — at its frontier leaves — it queries the CVPN for the counterfactual values of the subgame below, exactly as depth-limited solving uses a learned value at the horizon. The **expansion phase** then grows the tree: it simulates a single trajectory from the root, choosing actions by a PUCT rule that *mixes* the network’s prior with the current CFR policy, and appends the first public state it reaches that is not yet in the tree. Iterating — “expand the tree, improve the policy, expand, improve” — yields an *anytime* search that, like MCTS, pours computation into the relevant lines, but that, unlike MCTS, is solving for a game-theoretically sound strategy at every step.

The second idea is the **single knob that adapts this one search to both game classes**. When it expands, AlphaZero adds only the single most promising action, which is ideal when optimal play can be deterministic. But optimal imperfect-information play is generally *stochastic* — you must mix actions to avoid leaking information — so committing to one action is wrong. GT-CFR therefore expands the top- k actions by prior, and sets $k = 1$ for perfect-information games (one good line suffices; the search is then essentially AlphaZero’s) and $k = \infty$ — all children — for imperfect-information games (so the policy can mix over them). With $k = \infty$ the search even gains a *finite-time* guarantee on policy quality, not merely an in-the-limit one. This one switch is the technical heart of the unification: the same code reduces to

MCTS-like search on a chess position and to CFR-like iteration on a poker decision.

The third element is **sound self-play**, which trains the CVPN. Like AlphaZero, the agent is trained to predict the result of its own search: each network query made during a search defines a subgame, that subgame is re-solved by *another* GT-CFR search, and the network is trained toward the search’s (higher-quality) output — values by a Huber loss, policy by cross-entropy, with full-game outcomes supplying the remaining value targets. The word *sound* is the load-bearing one: every search used to generate data must be consistent with the network and with the other searches along the trajectory (it must not be assembling fragments of two different equilibria), which is enforced by running the searches on a safe re-solving auxiliary game. This is the imperfect-information analogue of AlphaZero’s bootstrapped self-play, and it is what makes the learned values actually converge to the game-theoretic ones.

The pairing comes with a guarantee that completes the chapter’s family of soundness bounds. Theorem 1 states that the regret of GT-CFR’s average policy after T iterations splits into a term that accumulates the value network’s ϵ -error over the tree’s *frontier* (the leaves the CVPN evaluates) and an ordinary CFR regret term over the tree’s *interior* (the nodes where CFR runs); dividing by T , the average policy’s exploitability is bounded by

$$\text{exploitability}(\bar{\pi}^T) \lesssim \underbrace{|\mathcal{F}| \epsilon}_{\text{value-net error (frontier)}} + \underbrace{\frac{|\mathcal{N}| UA}{\sqrt{T}}}_{\text{CFR convergence (interior)}},$$

with $|\mathcal{F}|$ and $|\mathcal{N}|$ the frontier and interior sizes, U the largest value gap, and A the maximum number of actions. The shape is by now familiar — a value-error term plus a $1/\sqrt{T}$ CFR-convergence term, the direct structural heir of DeepStack’s $k_1\epsilon + k_2/\sqrt{T}$ and ReBeL’s $\delta C_1 + \delta C_2/\sqrt{T}$ — but two things are distinctive. First, the coefficients are the *sizes of the growing tree’s frontier and interior*, so the bound is literally a statement about incremental tree growth: adding nodes over time costs nothing in convergence order. Second, and decisively, **this same statement holds whether the tree is a perfect-information game tree or an imperfect-information public tree** — it is the one soundness guarantee that spans both classes. A companion result, Theorem 2, then shows that invoking GT-CFR recursively through continual re-solving over a whole episode keeps the agent sound, with exploitability growing only *linearly* in the game length (a factor of roughly $5D + 2$ for D re-solving steps) — the property that lets Student of Games survive Scotland Yard’s twenty-four-round horizon. Together the theorems say what AlphaZero could only assume: as the network improves ($\epsilon \rightarrow 0$) and search deepens ($T \rightarrow \infty$), play converges to a Nash equilibrium — now for poker and chess alike.

6.6.4 Caveats, dead-ends, and what the paper under-describes

True to the chapter’s pattern, the *Science Advances* main text is the clean, high-level story and the engineering honesty lives in the supplement, the theorems being stated there “only informally” with the architectures, hyperparameters, full pseudocode, compute, and proofs all relegated to the Supplementary Text.

The headline caveat is openly owned: **Student of Games is weaker than AlphaZero in chess and Go given the same resources**, and the gap is not small in Go — its strongest configuration won just *2 of 400* games against a mid-training AlphaZero, i.e. it lost 99.5%, even while crushing the classical program Pachi by over 1,100 Elo. The authors call this “the price of SoG’s generality” and hypothesize the cause precisely: CFR is simply less efficient than Monte-Carlo tree search on perfect-information games, so the unified search pays for its breadth with peak strength. Schmid is candid that this is by design — the goal is one algorithm “better than humans” at everything, not the best chess engine — but it means the unification is a proof of *soundness and competence* across classes, not of dominance in either.

Two scaling walls are acknowledged as the real limits. The first is the **belief-space blow-up**, which Schmid names as the main limitation: the CVPN must *enumerate the information states per public state*, so its cost scales with the number of private states a player could occupy — manageable for poker’s 1,326 hands, but “prohibitively expensive in some games”, where the representation “falls apart”. This is exactly ReBeL’s PBS-input limitation inherited intact; the paper floats approximating it with a generative model that *samples* world states rather than enumerating them, but does not build one. The second is the **known-model requirement**: like AlphaZero and ReBeL, Student of Games needs a perfect simulator of the rules, placing it, in Schmid’s words, “behind the AlphaZero lineage” relative to MuZero, which *learns* its model.

Several smaller asterisks round out the picture. The system still keeps a **randomized betting abstraction** in poker (about twenty thousand actions reduced to four or five), the one piece of domain knowledge it has not shed, flagged as removable future work. Its search is **quadratic in the number of iterations** ($O(kT^2)$ network calls, since the tree grows each step), an inefficiency the authors note could be parallelized away but was not. The hyperparameter c — how many regret updates to run per expansion — is admittedly **under-explored** (“ $c = 1$ is not always best in practice ... we hope to explore this more thoroughly”). And the clean training-convergence argument is **idealized**, holding only “asymptotically, as $T \rightarrow \infty$ and with very large (exponential) memory”, so practice rests on the network’s finite capacity. None of these dents the result, but together they mark Student of Games as a first, deliberately general proof of concept rather than a tuned, scalable product.

6.6.5 Compute & accessibility

Student of Games’ cost is concentrated **offline, in TPU-scale self-play training**, and the paper reports it *relative to AlphaZero rather than in absolute terms* — a deliberate choice to enable a fair comparison without claiming a hardware record. The AlphaZero baseline used 3,500 concurrent actors each on a single Google TPUv4 over 800,000 training steps, and Student of Games “was trained using a similar amount of TPU resources”, with per-domain training running from roughly a million steps (Go) to seventeen million (Scotland Yard). Schmid’s own gloss is that chess and poker are trainable “on a few GPUs”, while Go “by far the hardest” consumed a large TPU pool — chiefly because the hardware was available, not because the method demanded it. There is, pointedly, **no single dollar figure** of the kind Pluribus made famous: because the comparison is pinned to *network-call budget* rather than wall-clock, the honest summary is “comparable to AlphaZero, and substantial” — indeed the authors list reducing this compute among their

open problems. At play time the search is anytime and tunable, but its $O(kT^2)$ growth (reducing to $O(T)$ for perfect-information games, where a node need only be evaluated once) means strong configurations are not cheap either.

On accessibility the verdict is mixed and worth stating precisely. The **full Student of Games agent and its trained networks were not released** — there is no code-availability statement beyond data in the paper and supplement — so on the headline-artefact axis it sits with the closed Libratus and Pluribus rather than the open ReBeL. What *is* open is the surrounding ecosystem: the same DeepMind/Alberta group maintains **OpenSpiel**, which ships the CFR family and the benchmark games themselves (Leduc, Liar’s Dice, Scotland Yard), so the components and environments are reproducible even though the assembled system is not. In practice this makes Student of Games an *open method on an open substrate with a closed flagship* — a reasonable middle ground, but one that leaves full replication of the chess/Go/poker/Scotland-Yard results beyond a typical group’s reach.

6.6.6 Strengths and limitations

Student of Games’ signal strength is **generality with soundness** — a combination no prior system in the chapter offered. It is the first algorithm to be *provably sound across both perfect- and imperfect-information games* (Theorems 1–2), and it backs the theory with breadth no competitor matched: in a single design it reaches expert-to-professional level in **chess and Go**, **beats Slumbot**, the strongest openly available heads-up no-limit bot (by about +7 mbb/g — the win-rate unit from the previous sections, which this paper writes as mbb/hand — and crucially it is *not* exploited by the local-best-response probe that catches weaker bots, which it beats by +434 mbb/g), and **defeats PimBot**, the state-of-the-art Scotland Yard agent built on Monte-Carlo tree search with hand-tuned heuristics, even when PimBot is given ten million search simulations to Student of Games’ few hundred. It uses **no human data, no precomputed blueprint, and no card abstraction**, learning everything from self-play; it **scales** with compute (more search and more training both lower exploitability, the empirical face of Theorem 1); and it demonstrably *refines its policy at test time through search*, “unlike any pure reinforcement-learning algorithm” — the clearest statement in the chapter of search-as-inference.

The limitations are the precise shape of its ambition. It is **weaker than the specialists** in their home domains — markedly so in Go — the acknowledged price of one algorithm for everything. Its guarantees, like ReBeL’s, hold only for **two-player zero-sum** play (Scotland Yard’s detectives are pooled into a single team to fit), so the *multiplayer* safety gap Pluribus exposed remains untouched even here. Its **belief representation does not scale** to games with vast private-state spaces, and it **requires a known model** of the environment, the two walls its own authors flag first. It retains a small **betting abstraction**, its search is **quadratic** in iterations, and — uniquely awkward for a generality claim — the **flagship code was never released**. The honest reading is that Student of Games proves the unification is *possible and sound*, and pays for that proof in peak strength, scale, and reproducibility.

6.6.7 Legacy and modern relevance

Student of Games is the capstone of this chapter’s argument, and it lands all three of its arcs at once. Across the five systems the field moved from **hand-built abstraction to learned neural approximation** (DeepStack and ReBeL dissolved the abstraction that Libratus and Pluribus depended on), and from **solving everything offline to learning-and-searching** (the blueprint shrank from Libratus’s petabytes to a value network and a real-time solve). Student of Games adds the third and final unification — **perfect and imperfect information in one algorithm** — and in doing so reframes the whole arc. The conceptual takeaway is that AlphaZero’s recipe (self-play, a learned value-and-policy network, and a tree grown by guided search) was never specific to perfect information; it only needed a *sound* search over *beliefs* rather than states. Swap Monte-Carlo tree search for GT-CFR and run it over public belief states, and the same recipe spans chess and poker with a single convergence guarantee. That is the chapter’s central thesis made literal: one algorithm, one network, one search, four very different games.

Its relationships to the surrounding frontier are clarifying. Student of Games is the *general* framework ReBeL pointed toward — it grows its tree instead of fixing a depth-limited subgame, decouples test-time search from training (so, like AlphaZero, it can simply search harder at test time for stronger play), and is validated across game classes rather than on imperfect information alone. Its most visible *unmet* neighbour is **MuZero**: where MuZero removed AlphaZero’s need for a known model by learning the dynamics, Student of Games still assumes a perfect simulator, so the natural next step — uniting GT-CFR with a learned model and a generative belief sampler — would dissolve both of its named limitations at once. And it sits one axis away from **CICERO**, Meta’s human-level Diplomacy player, which carries search-plus-learning into a seven-player, mixed-motive, natural-language game: a vivid reminder that Student of Games unifies *information structure* but not *player count*. Several of its parts are already reusable in their own right — **GT-CFR** as an anytime, policy-guided growing-tree search for any game class; the **CVPN** as a single value-and-policy head over belief states; **sound self-play** as a way to bootstrap targets from mutually consistent sub-searches; and a **domain-independent continual re-solving** that no longer leans on poker’s special structure.

For this thesis specifically, Student of Games is both a ceiling and a signpost. As the most general PBS-based sound-search framework, it is the natural context for **Contribution 1**: its belief — a distribution over the opponent’s possible information states — is exactly the substrate to be *enriched* with beliefs about the opponent’s strategy *type*, turning a Nash-seeking belief into an adaptive one. And its careful restriction of all guarantees to two-player zero-sum play — the paper is explicit that Nash is “less meaningful” beyond it — re-states, from the most advanced vantage point in the literature, the exact gap **Contribution 2** targets: unifying the *information structure* of games did nothing to close the *multi-agent safety* gap that Pluribus first exposed. That is the cleanest possible hand-off from this chapter to the rest of the dissertation, and the synthesis that follows draws the five systems together — an evolution-delta table of what each added and gave up, the evolution diagram, and the component-reuse map — to make the open problems, and the contributions that address them, explicit.

6.7 Synthesis

With all five systems in hand, the synthesis draws them together — not to crown a winner, but to make three things explicit: the shape of the seven-year arc, the machinery the systems share, and the open problems that hand this chapter off to the rest of the dissertation. The per-system scorecards already supply the dimension-by-dimension detail, so what follows aggregates rather than re-derives them.

6.7.1 The arc in one read

The progression is best read not as a climb but as a walk across the three axes of the introduction, in which every step forward on one axis was paid for somewhere else. DeepStack and Libratus opened the decade by attacking the same weakness — the lossy abstraction-and-blueprint paradigm — from opposite directions: DeepStack threw the paradigm out and replaced it with learned values and continual re-solving, while Libratus kept it and added a provably safe real-time repair. Both stayed two-player and neither reported the other’s headline result, so even at the start “better” was already multi-dimensional. Pluribus then advanced the *player-count* axis to six, the first superhuman result in any benchmark game beyond two players or teams, but did so by giving up safety entirely and remaining tabular — a sideways move on the representational axis and a backward one on guarantees. ReBeL reversed both of those concessions, returning to two players to *recover* a soundness guarantee while pushing the representational axis the furthest yet — public belief states with self-play-trained value and policy networks, no abstraction and no blueprint — and Student of Games completed the picture by adding the third axis, unifying perfect- and imperfect-information play in one algorithm, at the cost of peak strength in each. The honest one-line summary is that the field traded *hand-built breadth* for *learned generality* and *offline precomputation* for *real-time search*, and that no single system dominates on every axis at once.

The table below captures that walk as a ledger of what each system added and what it cost, which is the useful comparison for an evolutionary reading; the fixed per-dimension values (players, game type, blueprint, neural component, search, abstraction, perfect-information, compute, key innovation) are in each section’s scorecard and are not repeated here.

System (year)	What it added	What it gave up / what it cost
DeepStack (2017)	First <i>sound</i> depth-limited search under hidden information; learned counterfactual values in place of a stored strategy; discards the full-game abstraction-and-blueprint paradigm	Still keeps a sparse betting abstraction inside its look-ahead; never tested head-to-head against prior bots; ~175 CPU-core-years of offline value labelling

System (year)	What it added	What it gave up / what it cost
Libratus (2017/18)	First head-to-head <i>and</i> human win, via real-time nested safe subgame solving with a provable bound; an overnight self-improver that patches its own holes	Keeps and depends on abstraction + a petabyte blueprint; no neural generalization; supercomputer-scale (~25M core-hours)
Pluribus (2019)	First superhuman multiplayer (six-player) play; famously cheap (~\$150 on a single server)	Drops <i>all</i> safety guarantees (no N-player bound); relies on <i>unsafe</i> search; remains fully tabular and abstracted
ReBeL (2020)	“AlphaZero for imperfect information”: public belief states + learned value/policy + CFR in belief space; <i>recovers</i> the 2p0s guarantee; eliminates abstraction <i>and</i> blueprint; open-sources Liar’s Dice	Guarantees retreat to two players; the PBS blows up where common knowledge is scarce; GPU-cluster training; needs a known model
Student of Games (2023)	Unifies perfect- and imperfect-information play in one sound algorithm (GT-CFR + a single value-and-policy net + sound self-play)	Weaker than the specialists (markedly at Go); guarantees still 2p0s; belief-space blow-up; known-model requirement; flagship code unreleased

6.7.2 What carries forward

Almost nothing in these systems is wholly new; each is a recombination of a small set of building blocks, most of them introduced in Steps 3–5. **CFR and CFR** (Step 3) are the inner solver of every system; **Monte-Carlo CFR** (Step 3) trains the blueprints of Libratus and Pluribus; **card and action abstraction** (Step 4) underpin Libratus and Pluribus and survive only as a vestigial betting menu in ReBeL and Student of Games; and **neural value — then value-and-policy — approximation** (Step 5) is precisely what lets DeepStack, ReBeL, and Student of Games dissolve the abstraction the others depend on. Onto these the chapter layers its own native primitives, each introduced by one system and reused by its successors: **depth-limited solving**, **continual re-solving**, **safe nested subgame solving**, the **public belief state**, and **Growing-Tree CFR**. Read down the list of *first appearances* and the chapter’s evolution restates itself in components: the blocks accumulate, the abstraction-era ones fade, and the learned ones take over.

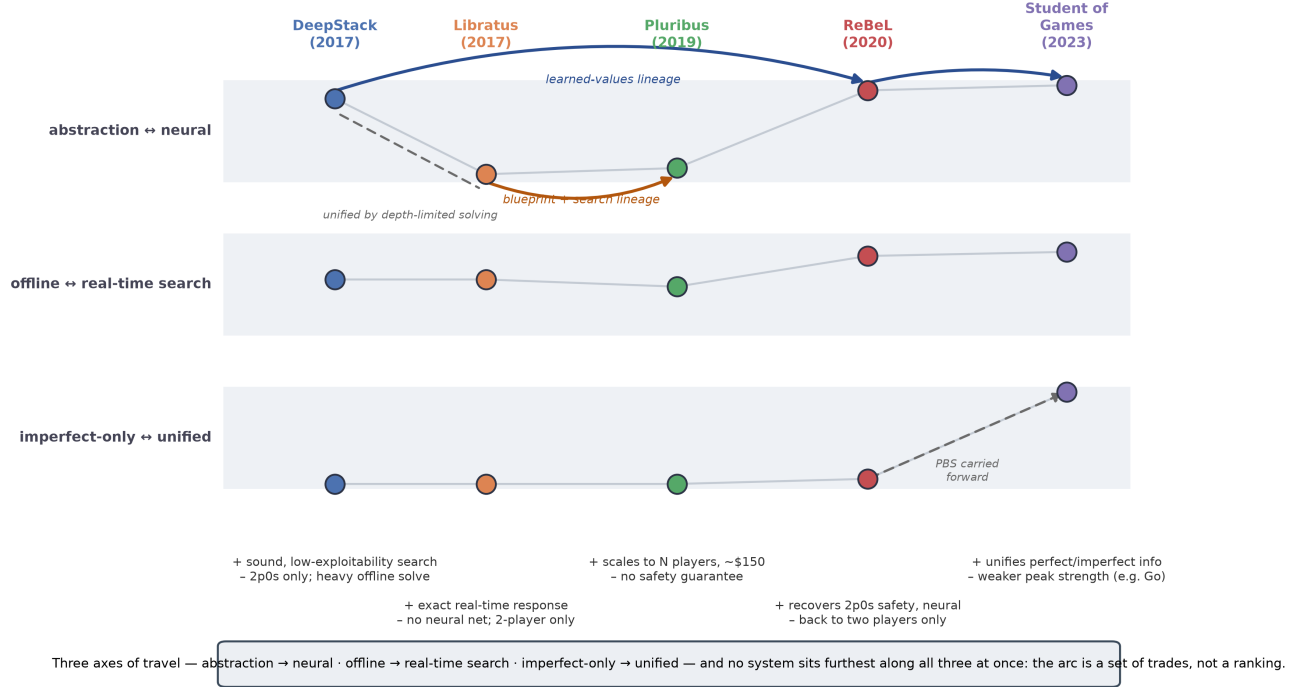


Figure 6: The seven-year arc as trade-offs: the five systems placed along three axes (abstraction → neural; offline → real-time search; imperfect-only → unified), with lineage arrows and a per-system capability-gained/given-up tag.

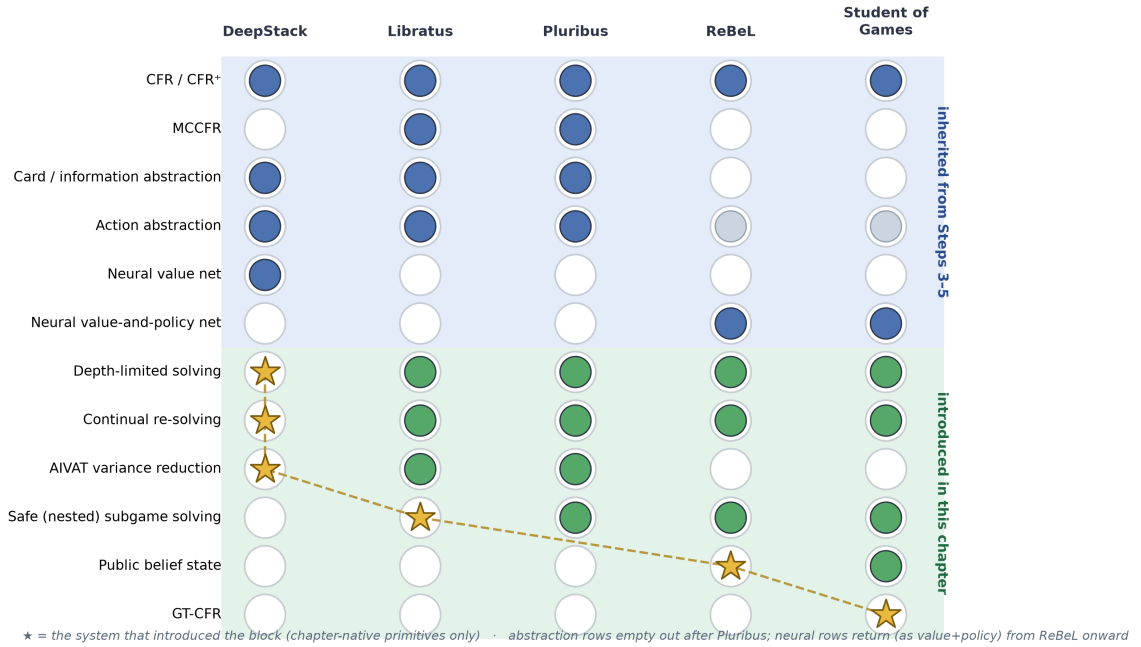


Figure 7: Component-reuse map: Step 3–5 building blocks and chapter-native primitives (rows) against the five systems in chronological order (columns); stars mark the system that introduced each native primitive, connected by a staircase of first appearances.

6.7.3 Why this matters for our research

This chapter is the state of the art the dissertation builds on and departs from, and its single most important contribution to that work is a framing one. Every system surveyed here is **opponent-blind by design**: each computes a worst-case-robust strategy and plays it unconditionally, and the two multiplayer- and human-tested systems make the stance explicit — they refuse to model or adapt to opponents so as never to be counter-exploited, and Pluribus does not even know who it is playing. Read as a survey, then, the chapter is a catalogue of five world-class systems that share exactly one omission: none adapts to the specific, fallible opponent in front of it. That shared omission is the opening this entire dissertation occupies, and the rest of this subsection reads the chapter’s assets against the three planned contributions.

For Contribution 1 (Behavioral Adaptation Framework). The chapter’s most direct gift is the **public belief state (PBS)** introduced by ReBeL and inherited by Student of Games — a distribution over the hidden states an opponent could occupy given common knowledge, which Contribution 1 widens to carry beliefs about the opponent’s *strategy type*, not merely their cards. Two less obvious points sharpen this. First, the PBS was invented to make *search sound*, and soundness is orthogonal to opponent modelling — so building adaptation on a PBS inherits the entire re-solving machinery and its convergence guarantees *unchanged*, letting an adaptive belief be updated inside a provably sound search rather than bolted on outside one; adaptation gets to reuse not just a data structure but the proofs attached to it. Second, every system in the chapter shares a **two-tier architecture** — a cheap global policy (a blueprint, or a learned value network) refined by a real-time local solve — and the dissertation’s own design, “start from a safe strategy and improve against weak opponents,” is that same two-tier shape with the second tier’s objective changed from *accuracy* to *bounded exploitation*. The chapter thus supplies both the substrate for adaptation and the system skeleton it slots into.

For Contribution 2 (Multi-Agent Safe Exploitation). The keystone is Pluribus: superhuman six-player poker reached with *no* safety guarantee of any kind, which is exactly the gap a theory of multi-agent safe exploitation must close — a point the chapter states outright. A subtler reading turns the systems’ own caveat into the contribution’s justification. Pluribus and Student of Games both restrict their guarantees to two-player zero-sum play and concede that beyond it a Nash equilibrium is neither unique, nor efficiently computable, nor even a guarantee against losing. If equilibrium buys no safety in the multiplayer setting to begin with, then declining to exploit a visibly weak opponent forfeits a guarantee one never possessed — which reframes deliberate exploitation from a reckless departure from safety into the *rational objective* once safety is unattainable by equilibrium alone. That is the cleanest motivation the literature offers for the multi-agent exploitation Contribution 2 pursues.

For Contribution 3 (Evaluation Methodology). The chapter standardizes on **exploitability** — the worst-case loss to a best-responding adversary — as the universal yardstick, applied identically to all five systems, and that uniformity is itself the template for a cross-system evaluation framework. It also hands the dissertation two instruments, repurposed. The **local-best-response (LBR) probe**, used throughout the chapter to *certify* unexploitability, becomes a *safety stress-test* when pointed at an adaptive agent — does chasing a weak opponent make you counter-exploitable? — which is precisely the adversarial,

teaching-attack-style evaluation Step 8 calls for. And **AIVAT**, the learned-value variance reducer behind DeepStack’s and Pluribus’s reported results, is the instrument that makes a small but *growing* adaptation edge statistically visible against the brutal variance of card games, where raw winnings would mislead.

Further leverage points (candidate — to keep or trim against final length). The following are compact extrapolations rather than claims already made by the systems; each is a one-line seed for later steps.

- **A reusable proof shape (C2).** Every guarantee in the chapter has the same form — a value-approximation term plus a $1/\sqrt{T}$ convergence term (DeepStack’s $k_1\epsilon + k_2/\sqrt{T}$, Libratus’s 2Δ , ReBeL’s $\delta C_1 + \delta C_2/\sqrt{T}$, SoG’s $|\mathcal{F}|\epsilon + |\mathcal{N}|UA/\sqrt{T}$) — giving Contribution 2 a ready template for bounding the exploitability *incurred* by a deliberate deviation.
- **Depth-limited exploitation (C1/C2).** Pluribus’s multi-valued leaves (k selectable continuation strategies) are structurally a menu of opponent behaviours at the horizon; reinterpreting that menu as opponent *types* yields tractable exploitation past the search depth — the exact problem Step 8’s “adapting beyond the depth limit” takes up.
- **A build-vs-cite map (methodology).** The “Compute & accessibility” subsections sort the field into what is reproducible at PhD scale (Pluribus’s single-server CFR, ReBeL’s single-thread CPU solver, the open OpenSpiel substrate, small games) versus what is cite-only (Libratus’s supercomputer, SoG’s TPU pool) — the implicit rationale for building on the cheap, open lineage (the ReBeL-Lite-on-Leduc task) rather than reproducing a flagship.
- **A redirected mechanism (framing).** Libratus’s overnight self-improvement is in-match strategy revision aimed at its *own* weaknesses; the dissertation runs the same observe-then-revise loop aimed at the *opponent’s*, so the contribution is redirecting a proven superhuman mechanism, not inventing one.
- **A frontier-narrative hook (framing).** All five systems are now cited as early instances of “test-time compute” — spending search at decision time rather than baking everything into a precomputed policy — so framing real-time adaptation as test-time *adaptation* compute connects this work to the contemporary reasoning-model story.
- **Consistency-as-safety (C2).** Student of Games’ “sound self-play” requires every local search to stay *consistent with* a safe reference, conceptually the same constraint as the KL-regularized (PiKL) exploitation Contribution 2 will use — an analogy worth borrowing.
- **Abstraction transferred to type-space (C3).** Step 4’s lesson that lossy abstraction creates exploitable seams carries from state-space to opponent-*type* space: a too-coarse type model is exploitable the way a too-coarse card abstraction is, predicting a failure mode for Contribution 3 to catch.

6.7.4 Open problems and the hand-off to Steps 7–15

Four open problems fall directly out of the synthesis, and each is the seed of a later step. The first is **opponent-blindness** itself: the entire frontier computes a fixed, worst-case-optimal strategy and plays it unconditionally, leaving the gap between static equilibrium and dynamic, opponent-aware play wide open — the subject of Step 7 (inferring opponent behaviour from action traces) and the inference half of

Contribution 1. The second is **multiplayer safety**: Pluribus proved that Nash-and-search methods *win* in N-player imperfect-information games while offering no safety guarantee at all, and Student of Games confirmed that even the most general sound-search framework still restricts its guarantees to two players — so a tractable theory of *safe exploitation beyond two-player zero-sum* (Step 8, Contribution 2) remains genuinely open. The third is **adaptation beyond the depth limit**: the chapter’s real-time solvers all look only a little way ahead, and carrying an opponent model soundly past that horizon — rather than discarding it at the leaf — is an unsolved problem Step 8 takes up directly. The fourth is **real-time compute budgets**: the systems’ costs range over six orders of magnitude, and an adaptive agent that must re-solve *and* re-estimate an opponent online inherits the tightest version of that constraint.

These problems define Phase D (Steps 7–8), the thesis-critical core, and they ramify through the phases that follow — multi-agent dynamics and coalition formation (Phase E), the data-driven behavioural pipelines that ground the models in real traces (Phase F), and the cross-domain evaluation framework that closes the plan (Phase G). The five systems of this chapter are, collectively, the platform this dissertation stands on and the foil it defines itself against: it inherits their belief-state representations, their real-time search, and their exploitability discipline, and it sets out to add the one capability they all, by design, leave out — the ability to notice that an opponent is not playing optimally, and to do something about it without becoming exploitable in turn.